



CAT 2021 Slot 3 Question Paper

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Instructions [1 - 4]

The passage below is accompanied by a set of questions. Choose the best answer to each question.

Starting in 1957, [Noam Chomsky] proclaimed a new doctrine: Language, that most human of all attributes, was innate. The grammatical faculty was built into the infant brain, and your average 3-year-old was not a mere apprentice in the great enterprise of absorbing English from his or her parents, but a "linguistic genius." Since this message was couched in terms of Chomskyan theoretical linguistics, in discourse so opaque that it was nearly incomprehensible even to some scholars, many people did not hear it. Now, in a brilliant, witty and altogether satisfying book, Mr. Chomsky's colleague Steven Pinker . . . has brought Mr. Chomsky's findings to everyman. In "The Language Instinct" he has gathered persuasive data from such diverse fields as cognitive neuroscience, developmental psychology and speech therapy to make his points, and when he disagrees with Mr. Chomsky he tells you so. . . .

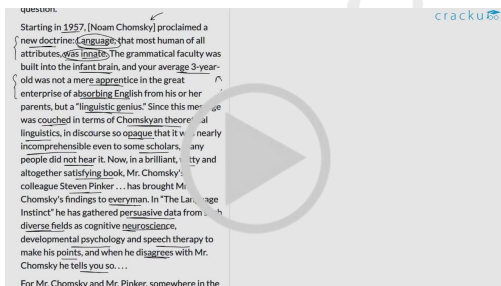
For Mr. Chomsky and Mr. Pinker, somewhere in the human brain there is a complex set of neural circuits that have been programmed with "super-rules" (making up what Mr. Chomsky calls "universal grammar"), and that these rules are unconscious and instinctive. A half-century ago, this would have been pooh-poohed as a "black box" theory, since one could not actually pinpoint this grammatical faculty in a specific part of the brain, or describe its functioning. But now things are different. Neurosurgeons [have now found that this] "black box" is situated in and around Broca's area, on the left side of the forebrain. . . .

Unlike Mr. Chomsky, Mr. Pinker firmly places the wiring of the brain for language within the framework of Darwinian natural selection and evolution. He effectively disposes of all claims that intelligent nonhuman primates like chimps have any abilities to learn and use language. It is not that chimps lack the vocal apparatus to speak; it is just that their brains are unable to produce or use grammar. On the other hand, the "language instinct," when it first appeared among our most distant hominid ancestors, must have given them a selective reproductive advantage over their competitors (including the ancestral chimps). . . .

So according to Mr. Pinker, the roots of language must be in the genes, but there cannot be a "grammar gene" any more than there can be a gene for the heart or any other complex body structure. This proposition will undoubtedly raise the hackles of some behavioral psychologists and anthropologists, for it apparently contradicts the liberal idea that human behavior may be changed for the better by improvements in culture and environment, and it might seem to invite the twin bugaboos of biological determinism and racism. Yet Mr. Pinker stresses one point that should allay such fears. Even though there are 4,000 to 6,000 languages today, they are all sufficiently alike to be considered one language by an extraterrestrial observer. In other words, most of the diversity of the world's cultures, so beloved to anthropologists, is superficial and minor compared to the similarities. Racial differences are literally only "skin deep." The fundamental unity of humanity is the theme of Mr. Chomsky's universal grammar, and of this exciting book.

1. Which one of the following statements best summarises the author's position about Pinker's book?

- A Culture and environment play a key role in shaping our acquisition of language.
- B Anatomical developments like the voice box play a key role in determining language acquisition skills.
- C The evolutionary and deterministic framework of Pinker's book makes it racist.
- D The universality of the "language instinct" counters claims that Pinker's book is racist.



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2. According to the passage, all of the following are true about the language instinct EXCEPT that:

- A all intelligent primates are gifted with it.
- B developments in neuroscience have increased its acceptance.
- C it confers an evolutionary reproductive advantage.
- D not all intelligent primates are gifted with it.

Starting in 1957, [Noam Chomsky] proclaimed a new doctrine: language that most human of all attributes is innate. The grammatical faculty was built into the infant brain, and your average 3-year-old was not a mere apprentice in the great enterprise of absorbing English from his or her parents, but a "linguistic genius." Since this message was couched in terms of Chomskyan theoretical linguistics, in discourse so opaque that it was nearly incomprehensible even to some scholars, any people did not hear it. Now, in a brilliant, witty and altogether satisfying book, Mr. Chomsky's colleague Steven Pinker... has brought Mr. Chomsky's findings to everyman. In "The Language Instinct" he has gathered persuasive data from such diverse fields as cognitive neuroscience, developmental psychology and speech therapy to make his points, and when he disagrees with Mr. Chomsky he tells you so....

For Mr. Chomsky and Mr. Pinker, somewhere in the

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3. On the basis of the information in the passage, Pinker and Chomsky may disagree with each other on which one of the following points?

- A The language instinct.
- B The inborn language acquisition skills of humans.
- C The Darwinian explanatory paradigm for language.
- D The possibility of a universal grammar.

Starting in 1957, [Noam Chomsky] proclaimed a new doctrine: language that most human of all attributes is innate. The grammatical faculty was built into the infant brain, and your average 3-year-old was not a mere apprentice in the great enterprise of absorbing English from his or her parents, but a "linguistic genius." Since this message was couched in terms of Chomskyan theoretical linguistics, in discourse so opaque that it was nearly incomprehensible even to some scholars, any people did not hear it. Now, in a brilliant, witty and altogether satisfying book, Mr. Chomsky's colleague Steven Pinker... has brought Mr. Chomsky's findings to everyman. In "The Language Instinct" he has gathered persuasive data from such diverse fields as cognitive neuroscience, developmental psychology and speech therapy to make his points, and when he disagrees with Mr. Chomsky he tells you so....

For Mr. Chomsky and Mr. Pinker, somewhere in the

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4. From the passage, it can be inferred that all of the following are true about Pinker's book, "The Language Instinct", EXCEPT that Pinker:

- A draws extensively from Chomsky's propositions.
- B disagrees with Chomsky on certain grounds.
- C draws from behavioural psychology theories.
- D writes in a different style from Chomsky.

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For Mr. Chomsky and Mr. Pinker, somewhere in the

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Instructions [5 - 8]

The passage below is accompanied by a set of questions. Choose the best answer to each question.

Today we can hardly conceive of ourselves without an unconscious. Yet between 1700 and 1900, this notion developed as a genuinely original thought. The "unconscious" burst the shell of conventional language, coined as it had been to embody the fleeting ideas and the shifting conceptions of several generations until, finally, it became fixed and defined in specialized terms within the realm of medical psychology and Freudian psychoanalysis.

The vocabulary concerning the soul and the mind increased enormously in the course of the nineteenth century. The enrichments of literary and intellectual language led to an altered understanding of the meanings that underlie time-honored expressions and traditional catchwords. At the same time, once coined, powerful new ideas attracted to themselves a whole host of seemingly unrelated issues, practices, and experiences, creating a peculiar network of preoccupations that as a group had not existed before. The drawn-out attempt to approach and define the unconscious brought together the spiritualist and the psychical researcher of borderline phenomena (such as apparitions, spectral illusions, haunted houses, mediums, trance, automatic writing); the psychiatrist or alienist probing the nature of mental disease, of abnormal ideation, hallucination, delirium, melancholia, mania; the surgeon performing operations with the aid of hypnotism; the magnetizer claiming to correct the disequilibrium in the universal flow of magnetic fluids but who soon came to be regarded as a clever manipulator of the imagination; the physiologist and the physician who puzzled over sleep, dreams, sleepwalking, anesthesia, the influence of the mind on the body in health and disease; the neurologist concerned with the functions of the brain and the physiological basis of mental life; the philosopher interested in the will, the emotions, consciousness, knowledge, imagination and the creative genius; and, last but not least, the psychologist.

Significantly, most if not all of these practices (for example, hypnotism in surgery or psychological magnetism) originated in the waning years of the eighteenth century and during the early decades of the nineteenth century, as did some of the disciplines (such as psychology and psychical research). The majority of topics too were either new or assumed hitherto unknown colors. Thus, before 1790, few if any spoke, in medical terms, of the affinity between creative genius and the hallucinations of the insane . . .

Striving vaguely and independently to give expression to a latent conception, various lines of thought can be brought together by some novel term. The new concept then serves as a kind of resting place or stocktaking in the development of ideas, giving satisfaction and a stimulus for further discussion or speculation. Thus, the massive introduction of the term unconscious by Hartmann in 1869 appeared to focalize many stray thoughts, affording a temporary feeling that a crucial step had been taken forward, a comprehensive knowledge gained, a knowledge that required only further elaboration, explication, and unfolding in order to bring in a bounty of higher understanding. Ultimately, Hartmann's attempt at defining the unconscious proved fruitless because he extended its reach into every realm of organic and inorganic, spiritual, intellectual, and instinctive existence, severely diluting the precision and compromising the impact of the concept.

5. Which one of the following statements best describes what the passage is about?

- A The discovery of the unconscious as a part of the human mind.
- B The growing vocabulary of the soul and the mind, as diverse processes.
- C The collating of diverse ideas under the single term: unconscious.
- D The identification of the unconscious as an object of psychical research.

The passage below is accompanied by a set of questions. Choose the best answer to each question.

Today we can hardly conceive of ourselves without an unconscious. Yet between 1700 and 1900 this notion developed as a genuinely original thought. The "unconscious" first the shell of conventional language, coined as it had been to embody fleeting ideas and the shifting conceptions; several generations until, finally, it became and defined in specialized terms within the realm of medical psychology and Freudian psychoanalysis.

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6. "The enrichments of literary and intellectual language led to an altered understanding of the meanings that underlie time-honored expressions and traditional catchwords." Which one of the following interpretations of this sentence would be closest in meaning to the original?

- A All of the options listed here.
- B Time-honored expressions and traditional catchwords were enriched by literary and intellectual language.
- C Literary and intellectual language was altered by time-honored expressions and traditional catchwords.
- D The meanings of time-honored expressions were changed by innovations in literary and intellectual language.

The passage below is accompanied by a set of questions. Choose the best answer to each question.

Today we can hardly conceive of ourselves without an unconscious. Yet between 1700 and 1900 this notion developed as a genuinely original thought. The "unconscious" first the shell of conventional language, coined as it had been to embody fleeting ideas and the shifting conceptions; several generations until, finally, it became and defined in specialized terms within the realm of medical psychology and Freudian psychoanalysis.

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7. Which one of the following sets of words is closest to mapping the main arguments of the passage?

- A Unconscious; Latent conception; Dreams.
- B Literary language; Unconscious; Insanity.

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Following statement

- infected by M

- B Tiny sentinels called X-Men infected people, turning them into mutant robot overlords.
- C None of the options listed here.
- D The X-Men were mutant heroes who now had to battle tiny robots called Nano-Sentinels

The passage below is accompanied by a set of questions. Choose the best answer to each question.

Back in the early 2000s, an awesome thing happened in the New X-Men comics. Our mutant heroes had been battling giant robots called Sentinels for years, but suddenly these mechanical overlords spawned a new threat: Nano-Sentinels! Not content to rule Earth with their metal fists, these tiny robots invaded our bodies at the microscopic level. Infected humans were slowly converted into machines, cell by cell.

Now, a new wave of extremely odd robots is making at least part of the Nano-Sentinels story come true. Using exotic fabrication materials like squishy hydrogels and elastic polymers, researchers are making autonomous devices that are often tiny and that could turn out to be more powerful than an army of Terminators. Some are 1-centimetre blobs

VIDEO SOLUTION

10. Which one of the following scenarios, if false, could be seen as supporting the passage?

- A Robots made from smart materials are likely to become part of our everyday lives in the future.
- B There are two kinds of molecules used to make some nano-robots: one that reacts positively to water and the other negatively.
- C Some hydrogels turn sticky when an electric current is passed through them; this potentially has very useful applications.
- D Nano-Sentinel-like robots are likely to be used to inject people to convert them into robots, cell by cell.

The passage below is accompanied by a set of questions. Choose the best answer to each question.

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VIDEO SOLUTION



11. Which one of the following statements, if true, would be the most direct extension of the arguments in the passage?

- A Sentinel robots will be used in warfare to cause large-scale destructive mutations amongst civilians.
- B X-Men may be created by injecting people with mutant nano-gels that will respond to the brain's magnetic field.
- C In the future, robots will be used to search and destroy diseases even in the deepest recesses of the human body.
- D 1-centimetre blobs of gel that have nano-robots in them will be used to send messages.

The passage below is accompanied by a set of questions. Choose the best answer to each question.

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VIDEO SOLUTION

12. Which one of the following statements best summarises the central point of the passage?

- A Robots will use nano-robots on their feet and wheels to climb walls or move on ceilings.

- The passage below is accompanied by a set of questions. Choose the best answer to each question.
- Back in the early 2000s, an awesome thing happened in the New X-Men comics. Our mutant heroes had been battling giant robots called Sentinels for years, but suddenly these mechanical overlords spawned a new threat: Nano-Sentinels. Not content to rule Earth with their metal fists, these tiny robots invaded our bodies at the microscopic level. Infected humans were converted into machines, cell by cell.
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Instructions [13 - 16]

Keeping time accurately comes with a price. The maximum accuracy of a clock is directly related to how much disorder, or entropy, it creates every time it ticks. Natalia Ares at the University of Oxford and her colleagues made this discovery using a tiny clock with an accuracy that can be controlled. The clock consists of a 50-nanometre-thick membrane of silicon nitride, vibrated by an electric current. Each time the membrane moved up and down once and then returned to its original position, the researchers counted a tick, and the regularity of the spacing between the ticks represented the accuracy of the clock. The researchers found that as they increased the clock's accuracy, the heat produced in the system grew, increasing the entropy of its surroundings by jostling nearby particles. . . "If a clock is more accurate, you are paying for it somehow," says Ares. In this case, you pay for it by pouring more ordered energy into the clock, which is then converted into entropy. "By measuring time, we are increasing the entropy of the universe," says Ares. The more entropy there is in the universe, the closer it may be to its eventual demise. "Maybe we should stop measuring time," says Ares. The scale of the additional entropy is so small, though, that there is no need to worry about its effects, she says.

The relationship that the researchers found is a limit on the accuracy of a clock, so it doesn't mean that a clock that creates the most possible entropy would be maximally accurate - hence a large, inefficient grandfather clock isn't more precise than an atomic clock. "It's a bit like fuel use in a car. Just because I'm using more fuel doesn't mean that I'm going faster or further," says Huber.

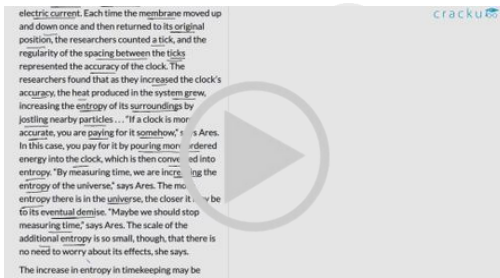
13. None of the following statements can be inferred from the passage EXCEPT that:

- electric current. Each time the membrane moved up and down once and then returned to its original position, the researchers counted a tick, and the regularity of the spacing between the ticks represented the accuracy of the clock. The researchers found that as they increased the clock's accuracy, the heat produced in the system grew, increasing the entropy of its surroundings by jostling nearby particles. "If a clock is more accurate, you are paying for it somehow," says Ares. In this case, you pay for it by pouring more energy into the clock, which is then converted into entropy. "By measuring time, we are increasing the entropy of the universe," says Ares. The more entropy there is in the universe, the closer it is to its eventual demise. "Maybe we should stop measuring time," says Ares. The scale of the additional entropy is so small, though, that there is no need to worry about its effects, she says. The increase in entropy in timekeeping may be

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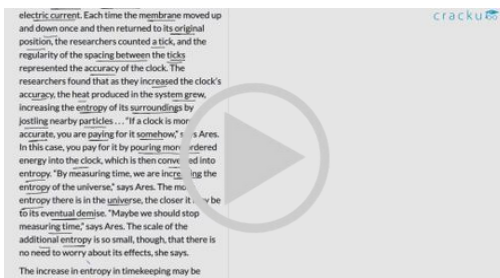
- A The relationship between accuracy and entropy may not apply to all clocks.
- B Researchers found that the heat produced in a system is the price paid for increased accuracy of measurement.
- C There is no difference in accuracy between an inefficient grandfather clock and an atomic clock.
- D In designing clocks for quantum computers, both precision and heat have to be taken into account.



VIDEO SOLUTION

15. "It's a bit like fuel use in a car. Just because I'm using more fuel doesn't mean that I'm going faster or further . . ." What is the purpose of this example?

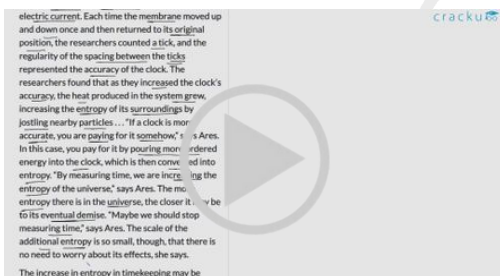
- A If you go faster in a car, you will tend to consume more fuel, but the converse is not necessarily true. In the same way, increased entropy does not necessarily mean greater accuracy of a clock.
- B The further you go in a car, the more fuel you use. In the same way, the faster you go in a car, the less time you use.
- C If you measure the speed of a car with a grandfather clock, the result will be different than if you measured it with an atomic clock.
- D The further and faster you go in a car, the greater the amount of fuel you will use, the greater the amount of heat produced and, hence, the greater the entropy.



VIDEO SOLUTION

16. Which one of the following sets of words and phrases serves best as keywords of the passage?

- A Electric current; Heat; Quantum effects.
- B Silicon Nitride; Energy; Grandfather Clock.
- C Measuring Time; Accuracy; Entropy.
- D Membrane; Arrow of time; Entropy.



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17. The four sentences (labelled 1, 2, 3 and 4) below, when properly sequenced would yield a coherent paragraph. Decide on the proper sequencing of the order of the sentences and key in the sequence of the four numbers as your answer:

1. Businesses find automation, such as robotic employees, a big asset in terms of productivity and efficiency.
2. But in recent years, robotics has had increasing impacts on unemployment, not just of manual labour, as computers are rapidly handling some white-collar and service-sector work.
3. For years politicians have promised workers that they would bring back their jobs by clamping down on trade, offshoring and immigration.
4. Economists, based on their research, say that the bigger threat to jobs now is not globalisation but automation.

The four sentences (labelled 1, 2, 3 and 4) below, when properly sequenced would yield a coherent paragraph. Decide on the proper sequencing of the order of the sentences and key in the sequence of the four numbers as your answer:

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4. Economists, based on their research, say that the bigger threat to jobs now is not globalisation but automation.

Answer: _____

A  B

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18. The four sentences (labelled 1, 2, 3, 4) below, when properly sequenced would yield a coherent paragraph. Decide on the proper sequencing of the order of the sentences and key in the sequence of the four numbers as your answer:

1. It is regimes of truth that make certain relationships speakable - relationships, like subjectivities, are constituted through discursive formations, which sustain regimes of truth.
2. Relationships are nothing without the communication that brings them into being; interpersonal communication is connected to knowledge shared by interlocutors, and scholars should attend to relational histories in their analyses.
3. A Foucauldian approach to relationships goes beyond these conceptions of discourse and history to macrolevel regimes of truth as constituting relationships.
4. Reconsidering micropractices within relationships that are constituted within and simultaneously contributors to regimes of truth acknowledges the central position of power/knowledge in the constitution of what has come to be considered true and real.

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Answer: _____

A  B

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19. Five jumbled up sentences, related to a topic, are given below. Four of them can be put together to form a coherent paragraph. Identify the odd one out and key in the number of the sentence as your answer:

1. A typical example is Wikipedia, where the overwhelming majority of contributors are male and so the available content is skewed to reflect their interests.
2. Without diversity of thought and representation, society is left with a distorted picture of future options, which are likely to result in augmenting existing inequalities.
3. Gross gender inequality in the technology sector is problematic, not only for the industry-wide marginalisation of women, but because technology designs embody the values of their makers.
4. While redressing unequal representation in the workplace is a step in the right direction, broader social change is needed to address the structural inequalities embedded within the current organisation of work and employment.
5. If technology merely reflects the perspectives of the male stereotype, then new technologies are unlikely to accommodate the diverse social contexts within which they operate.

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Answer: _____

A  B

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20. The passage given below is followed by four alternate summaries. Choose the option that best captures the essence of the passage.

People view idleness as a sin and industriousness as a virtue, and in the process have developed an unsatisfactory relationship with their jobs. Work has become a way for them to keep busy, even though many find their work meaningless. In their need for activity people undertake what was once considered work (fishing, gardening) as hobbies. The opposing view is that hard work has made us prosperous and improved our levels of health and education. It has also brought innovation and labour and time-saving devices, which have lessened life's drudgery.

- A Despite some detractors, hard work is essential in today's world to enable economic progress, for education and health and to propel innovations that make life easier.
- B Hard work has overtaken all aspects of our lives and has enabled economic prosperity, but it is important that people reserve their leisure time for some idleness.
- C Some believe that hard work has been glorified to the extent that it has become meaningless, and led to greater idleness, but it has also had enormous positive impacts on everyday life.
- D While the idealisation of hard work has propelled people into meaningless jobs and endless activity, it has also led to tremendous social benefits from prosperity and innovation.

The passage given below is followed by four alternate summaries. Choose the option that best captures the essence of the passage.

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A) Despite some detractors, hard work is essential to enable economic progress, for education and health and to propel innovations that make life easier.

B) Hard work has overtaken all aspects of our lives and has enabled economic prosperity, but it is important that people reserve their leisure time for some idleness.

C) Some believe that hard work has been glorified to the extent that it has become meaningless, and led to greater idleness, but it has also had enormous positive impacts on everyday life.

D) While the idealisation of hard work has propelled people into meaningless jobs and endless activity, it has also led to tremendous social benefits from prosperity and innovation.

21. Five jumbled up sentences, related to a topic, are given below. Four of them can be put together to form a coherent paragraph. Identify the odd one out and key in the number of the sentence as your answer:

1. They often include a foundation course on navigating capitalism with Chinese characteristics and have replaced typical cases from US corporates with a focus on how Western theories apply to China's buzzing local firms.
2. The best Chinese business schools look like their Western rivals but are now growing distinct in terms of what they teach and the career boost they offer.
3. Western schools have enhanced their offerings with double degrees, popular with domestic and overseas students alike—and boosted the prestige of their Chinese partners.
4. For students, a big draw is the chance to rub shoulders with captains of China's private sector.
5. Their business courses now largely cater to the growing demand from China Inc which has become more global, richer and ready to recruit from this sinocentric student body.

Five jumbled up sentences, related to a topic, are given below. Four of them can be put together to form a coherent paragraph. Identify the odd one out and key in the number of the sentence as your answer:

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Answer: _____

22. The passage given below is followed by four alternate summaries. Choose the option that best captures the essence of the passage.

The human mind is wired to see patterns. Not only does the brain process information as it comes in, it also stores insights from all our past experiences. Every interaction, happy or sad, is catalogued in our memory. Intuition draws from that deep memory well to inform our decisions going forward. In other words, intuitive decisions are based on data, and not contrary to data as many would like to assume. When we subconsciously spot patterns, the body starts firing neurochemicals in both the brain and gut. These "somatic markers" are what give us that instant sense that something is right ... or that it's off. Not only are these automatic processes faster than rational thought, but our intuition draws from decades of diverse qualitative experience (sights, sounds, interactions, etc.) - a wholly human feature that big data alone could never accomplish.

- A Intuition is infinitely richer than big data which is based on rational thought and accomplishes more than what big data can.

- B Intuitions are automatic processes and are therefore faster than rational thought, and so decisions based on them are better.
- C Intuition draws from deep memory, and may not be related to data, but to decades of diverse qualitative experience.
- D Intuitions are neuro-chemical firings based on pattern recognition and draw upon a rich and vast database of experiences.

The passage given below is followed by four alternate summaries. Choose the option that best captures the essence of the passage.

The human mind is wired to see patterns. Not only does the brain process information as it comes in, it also stores insights from all our past experiences. Every interaction, happy or sad, is catalogued in our memory. Intuition draws from that deep memory well to inform our decisions going forward. In other words, intuitive decisions are based on data, and not contrary to data as many would like to assume. When we subconsciously spot patterns, the body starts firing neurochemicals in both the brain and gut. These "somatic markers" are what give us that instant sense that something is right... or that it's off. Not only do intuitive processes faster than rational thought, but our intuition draws from decades of diverse qualitative experience – sights, sounds, interactions – a wholly human feature that big data alone could never accomplish.

A) Intuition is infinitely richer than big data which is based on rational thought and accomplishes more than what big data can. Intuitions are automatic processes and are therefore faster than rational thought, and so decisions based on them are better.

C) Intuition draws from deep memory and not data, but to decades of diverse qualitative experience. Intuitions are neuro-chemical firings based on pattern recognition and draw upon a rich and vast database of experiences.

VIDEO SOLUTION



23. The passage given below is followed by four alternate summaries. Choose the option that best captures the essence of the passage.

Brazil's growth rate has been low, yet most Brazilians say their financial situation has improved, and they expect it to get even better. This is because most incomes are rising fast, with higher minimum wages and very low unemployment. The result is falling inequality and a growing middle class – the result of economic stabilization, improved social security and universal primary education. But despite recent improvements the Brazilian economy is still painfully unequal, with poor Brazilians paying the biggest share of their income in taxes and getting the least back in government services.

- A Economic reforms have benefitted many Brazilians, but they are unaware of the impending problems from rising inequalities in their society.
- B Good economic indicators have masked the unfair taxation of the poor that is likely to destabilise the Brazilian economy in the next few years.
- C Most Brazilians feel they have benefitted from recent economic events, but the poor continue to be dealt unfairly by the state.
- D With rising incomes and falling unemployment, most Brazilians are being misled into thinking that their economy is doing well.

The passage given below is followed by four alternate summaries. Choose the option that best captures the essence of the passage.

Brazil's growth rate has been low, yet most Brazilians say their financial situation has improved, and they expect it to get even better. This is because most incomes are rising fast, with higher minimum wages and very low unemployment. The result is falling inequality and a growing middle class – the result of economic stabilization, improved social security and universal primary education. But despite recent improvements the Brazilian economy is still painfully unequal, with poor Brazilians paying the biggest share of their income in taxes and getting the least back in government services.

A) Economic reforms have benefitted many Brazilians, but they are unaware of the impending problems from rising inequalities in their society.

B) Good economic indicators have masked the unfair taxation of the poor that is likely to destabilise the Brazilian economy in the next few years.

C) Most Brazilians feel they have benefitted from recent economic events, but the poor continue to be dealt unfairly by the state.

D) With rising incomes and falling unemployment, most Brazilians are being misled into thinking that their economy is doing well.

VIDEO SOLUTION

24. The four sentences (labelled 1, 2, 3 and 4) below, when properly sequenced would yield a coherent paragraph. Decide on the proper sequencing of the order of the sentences and key in the sequence of the four numbers as your answer:

1. Restitution of artefacts to original cultures could face legal obstacles, as many Western museums are legally prohibited from disposing off their collections.
2. This is in response to countries like Nigeria, which are pressurising European museums to return their precious artefacts looted by colonisers in the past.
3. Museums in Europe today are struggling to come to terms with their colonial legacy, some taking steps to return artefacts but not wanting to lose their prized collections.
4. Legal hurdles notwithstanding, politicians and institutions in France and Germany would now like to defuse the colonial time bombs, and are now backing the return of part of their holdings.

The four sentences (labelled 1, 2, 3 and 4) below, when properly sequenced would yield a coherent paragraph. Decide on the proper sequencing of the order of the sentences and key in the sequence of the four numbers as your answer.

1. Restitution of artefacts to original cultures could face legal obstacles, as many Western museums are legally prohibited from disposing off their collections.
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3. Museums in Europe today are struggling to come to terms with their colonial legacy, some taking steps to return artefacts but not wanting to lose their prized collections.
4. Legal hurdles notwithstanding, politicians and institutions in France and Germany would now like to defuse the colonial time bombs, and are now backing the return of part of their holdings.

Answer: _____



LRDI

Instructions [25 - 28]

Each of the bottles mentioned in this question contains 50 ml of liquid. The liquid in any bottle can be 100% pure content (P) or can have certain amount of impurity (I). Visually it is not possible to distinguish between P and I. There is a testing device which detects impurity, as long as the percentage of impurity in the content tested is 10% or more.

For example, suppose bottle 1 contains only P, and bottle 2 contains 80% P and 20% I. If content from bottle 1 is tested, it will be found out that it contains only P. If content of bottle 2 is tested, the test will reveal that it contains some amount of I. If 10 ml of content from bottle 1 is mixed with 20 ml content from bottle 2, the test will show that the mixture has impurity, and hence we can conclude that at least one of the two bottles has I. However, if 10 ml of content from bottle 1 is mixed with 5 ml of content from bottle 2, the test will not detect any impurity in the resultant mixture.

25. 5 ml of content from bottle A is mixed with 5 ml of content from bottle B. The resultant mixture, when tested, detects the presence of I. If it is known that bottle A contains only P, what BEST can be concluded about the volume of I in bottle B?

- A 1 ml
- B Less than 1 ml
- C 10 ml
- D 10 ml or more

Each of the bottles mentioned in this question contains 50 ml of liquid. The liquid in any bottle can be 100% pure content (P) or can have certain amount of impurity (I). Visually it is not possible to distinguish between P and I. There is a testing device which detects impurity, as long as the percentage of impurity in the content tested is 10% or more.

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Question (25)

5 ml of content from bottle A is mixed with 5 ml of content from bottle B. The resultant mixture, when tested, detects the presence of I. It is known that bottle A contains only P. What BEST can be concluded about the volume of I in bottle B?

A) 1 ml
B) Less than 1 ml
C) 10 ml
D) 10 ml or more

Handwritten notes: (A) 10%, (B) 20%, volume of I = ?



26. There are four bottles. Each bottle is known to contain only P or only I. They will be considered to be "collectively ready for despatch" if all of them contain only P. In minimum how many tests, is it possible to ascertain whether these four bottles are "collectively ready for despatch"?

Each of the bottles mentioned in this question contains 50 ml of liquid. The liquid in any bottle can be 100% pure content (P) or can have certain amount of impurity (I). Visually it is not possible to distinguish between P and I. There is a testing device which detects impurity, as long as the percentage of impurity in the content tested is 10% or more.

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Question (26)

There are four bottles. Each bottle is known to contain only P or only I. They will be considered to be "collectively ready for despatch" if all of them contain only P. In minimum how many tests, is it possible to ascertain whether these four bottles are "collectively ready for despatch"?

A) 1
B) 2
C) 3
D) 4

Handwritten notes: (A) 10%, (B) 20%, volume of I = ?

27. There are four bottles. It is known that three of these bottles contain only P, while the remaining one contains 80% P and 20% I. What is the minimum number of tests required to definitely identify the bottle containing some amount of I?

cracku

Each of the bottles mentioned in this question contains 20 ml of liquid. The liquid in any bottle can be 100% pure content (P) or can have certain amount of impurity (I). Visually it is not possible to distinguish between P and I. There is a testing device which detects impurity, as long as the percentage of impurity in the content tested is 10% or more.

For example, suppose bottle 1 contains only P and bottle 2 contains 80% P and 20% I. If content from bottle 1 is tested, it will be found out that it contains only P. If content of bottle 2 is tested, the test will reveal that it contains some amount of I. If 10 ml of content from bottle 1 is mixed with 10 ml content from bottle 2, the test will show that the mixture is impure, and hence we can conclude that at least one of the bottles has I. However, if 10 ml of content from bottle 1 is mixed with 5 ml content from bottle 2, the test will not detect any impurity in the mixture.

Question (1)

5 ml of eggplant from bottle A is mixed with 5 ml of eggplant from bottle B. The resultant mixture, when tested, detects the presence of I. It is known that bottle A contains only P, what BEST can be concluded about the volume of I in bottle B?

A) 1 ml
B) Less than 1 ml
C) 10 ml or more

Handwritten notes: (A) 10%, (B) 20%, volume of I

VIDEO SOLUTION

28. There are four bottles. It is known that either one or two of these bottles contain(s) only P, while the remaining ones contain 85% P and 15% I. What is the minimum number of tests required to ascertain the exact number of bottles containing only P?

- A 4
B 2
C 3
D 1

cracku

Each of the bottles mentioned in this question contains 20 ml of liquid. The liquid in any bottle can be 100% pure content (P) or can have certain amount of impurity (I). Visually it is not possible to distinguish between P and I. There is a testing device which detects impurity, as long as the percentage of impurity in the content tested is 10% or more.

For example, suppose bottle 1 contains only P and bottle 2 contains 80% P and 20% I. If content from bottle 1 is tested, it will be found out that it contains only P. If content of bottle 2 is tested, the test will reveal that it contains some amount of I. If 10 ml of content from bottle 1 is mixed with 10 ml content from bottle 2, the test will show that the mixture is impure, and hence we can conclude that at least one of the bottles has I. However, if 10 ml of content from bottle 1 is mixed with 5 ml content from bottle 2, the test will not detect any impurity in the mixture.

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5 ml of eggplant from bottle A is mixed with 5 ml of eggplant from bottle B. The resultant mixture, when tested, detects the presence of I. It is known that bottle A contains only P, what BEST can be concluded about the volume of I in bottle B?

A) 1 ml
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Handwritten notes: (A) 10%, (B) 20%, volume of I

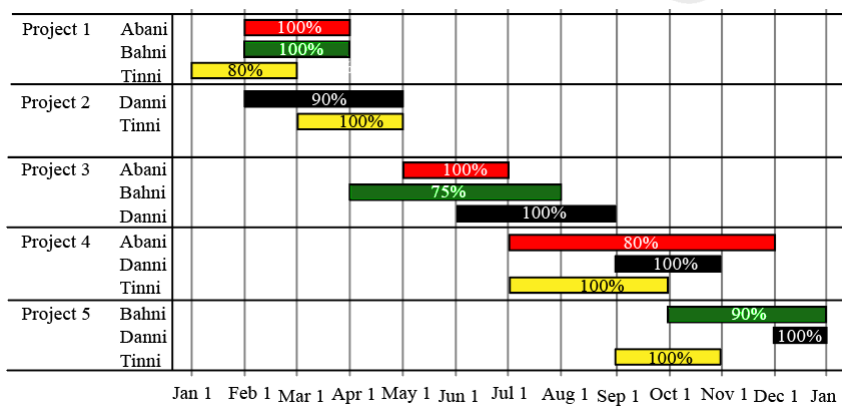
VIDEO SOLUTION

cracku

CAT Daily Targets

Step by step, heading towards 99+ percentile

Instructions [29 - 32]



The figure above shows the schedule of four employees - Abani, Bahni, Danni, and Tinni - whom Dhoni supervised in 2020. Altogether there were five projects which started and concluded in 2020 in which they were involved. For each of these projects and for each employee, the starting day was at the beginning of a month and the concluding day was the end of a month, and these are indicated by the left and right end points of the corresponding horizontal bars. The number within each bar indicates the percentage of assigned work completed by the employee for that project, as assessed by Dhoni.

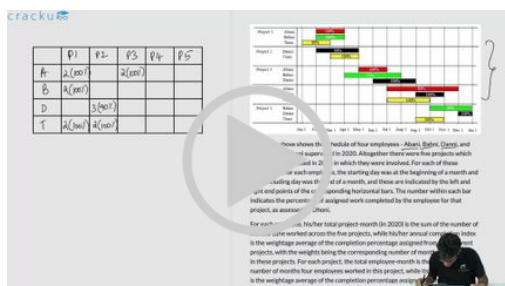
For each employee, his/her total project-month (in 2020) is the sum of the number of months (s)he worked across the five projects, while his/her annual completion index is the weightage average of the completion percentage assigned from the different projects, with the weights being the corresponding number of months (s)he worked in these projects. For each project, the total employee-month is the sum of the number of months four employees worked in this project, while its completion index is the weightage average of the completion percentage assigned for the employees who worked in this project, with the weights being the corresponding number of months they worked in this project.

29. Which of the following statements is/are true?

- I: The total project-month was the same for the four employees.
II: The total employee-month was the same for the five projects.

A Only II

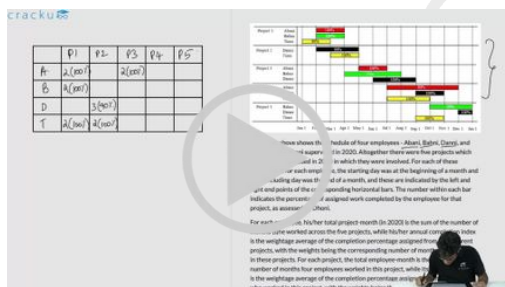
- B Both I and II
- C Neither I nor II
- D Only I



VIDEO SOLUTION

30. Which employees did not work in multiple projects for any of the months in 2020?

- A Only Abani, Bahni and Danni
- B Only Abani and Bahni
- C All four of them
- D Only Tinni



VIDEO SOLUTION

31. The project duration, measured in terms of the number of months, is the time during which at least one employee worked in the project. Which of the following pairs of the projects had the same duration?

- A Project 1, Project 2
- B Project 4, Project 5
- C Project 3, Project 5
- D Project 3, Project 4



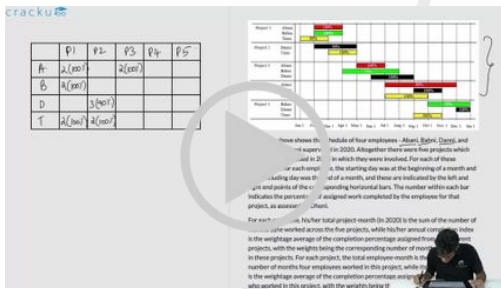
VIDEO SOLUTION

FREE

CAT Sectional Tests

32. The list of employees in decreasing order of annual completion index is:

- A Danni, Tinni, Bahni, Abani
 B Bahni, Abani, Tinni, Danni
 C Danni, Tinni, Abani, Bahni
 D Tinni, Danni, Abani, Bahni



VIDEO SOLUTION

Instructions [33 - 38]

10 players - P1, P2, ..., P10 - competed in an international javelin throw event. The number (after P) of a player reflects his rank at the beginning of the event, with rank 1 going to the topmost player. There were two phases in the event with the first phase consisting of rounds 1, 2, and 3, and the second phase consisting of rounds 4, 5, and 6. A throw is measured in terms of the distance it covers (in meters, up to one decimal point accuracy), only if the throw is a 'valid' one. For an invalid throw, the distance is taken as zero. A player's score at the end of a round is the maximum distance of all his throws up to that round. Players are re-ranked after every round based on their current scores. In case of a tie in scores, the player with a prevailing higher rank retains the higher rank. This ranking determines the order in which the players go for their throws in the next round.

In each of the rounds in the first phase, the players throw in increasing order of their latest rank, i.e. the player ranked 1 at that point throws first, followed by the player ranked 2 at that point and so on. The top six players at the end of the first phase qualify for the second phase. In each of the rounds in the second phase, the players throw in decreasing order of their latest rank i.e. the player ranked 6 at that point throws first, followed by the player ranked 5 at that point and so on. The players ranked 1, 2, and 3 at the end of the sixth round receive gold, silver, and bronze medals respectively.

All the valid throws of the event were of distinct distances (as per stated measurement accuracy). The tables below show distances (in meters) covered by all valid throws in the first and the third round in the event.

Distances covered by all the valid throws in the first round

Player	Distance(in m)
P1	82.9
P3	81.5
P5	86.4
P6	82.5
P7	87.2
P9	84.1

Distances covered by all the valid throws in the third round

Player	Distance(in m)
P1	88.6
P3	79.0
P9	81.4

The following facts are also known.

- Among the throws in the second round, only the last two were valid. Both the throws enabled these players to qualify for the second phase, with one of them qualifying with the least score. None of these players won any medal.
- If a player throws first in a round AND he was also the last (among the players in the current round) to throw in the previous round, then the player is said to get a double. Two players got a double.
- In each round of the second phase, exactly one player improved his score. Each of these improvements was by the same amount.
- The gold and bronze medalists improved their scores in the fifth and the sixth rounds respectively. One medal winner improved his score in the fourth round.
- The difference between the final scores of the gold medalist and the silver medalist, as well as the difference between the final scores of the silver medalist and the bronze medalist was 1.0 m.

33. Which two players got the double?

- A P1, P8
 B P2, P4
 C P8, P10
 D P1, P10

Handwritten notes on the left margin:

1. P1, P2, P3, P4, P5, P6, P7, P8, P9, P10

Handwritten notes on the right margin:

order

qualify for the second phase. In each of the rounds in the second phase, the players throw in decreasing order of their latest rank i.e. the player ranked 6 at that point throw first, followed by the player ranked 5 at that point and so on. The players ranked 1, 2, and 3 at the end of the sixth round receive gold, silver, and bronze medals respectively.

All the valid throws of the event were of distinct distances (as per stated measurement accuracy). The tables below show distances (in meters) covered by all valid throws in the first and the third round in the event.

Distances covered by all valid throws in the first round

Player	Distance (m)
P1	82.5
P2	81.5
P3	80.4
P4	82.7
P5	87.2
P6	81.1

Distances covered by all valid throws in the third round

Player	Distance (m)
P1	85.0
P2	79
P3	83.8

The following are also known:

- Among the throws in the second round, only the last two were valid. Both throws enabled these players to qualify for the second phase, with one of them qualifying with the least score. None of these players won any medal.
- If a player throws first in a round AND he was also the last (among the current round) to throw in the previous round, then the players get a double. Two players got a double.

VIDEO SOLUTION

34. Who won the silver medal?

- A P5
- B P7
- C P9
- D P1

Handwritten notes on the left margin:

1. P1, P2, P3, P4, P5, P6, P7, P8, P9, P10

Handwritten notes on the right margin:

order

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P1	85.0
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The following are also known:

- Among the throws in the second round, only the last two were valid. Both throws enabled these players to qualify for the second phase, with one of them qualifying with the least score. None of these players won any medal.
- If a player throws first in a round AND he was also the last (among the current round) to throw in the previous round, then the players get a double. Two players got a double.

VIDEO SOLUTION



35. Who threw the last javelin in the event?

- A P7
- B P1
- C P9
- D P10

Handwritten notes on the left margin:

1. P1, P2, P3, P4, P5, P6, P7, P8, P9, P10

Handwritten notes on the right margin:

order

qualify for the second phase. In each of the rounds in the second phase, the players throw in decreasing order of their latest rank i.e. the player ranked 6 at that point throw first, followed by the player ranked 5 at that point and so on. The players ranked 1, 2, and 3 at the end of the sixth round receive gold, silver, and bronze medals respectively.

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P5	87.2
P6	81.1

Distances covered by all valid throws in the third round

Player	Distance (m)
P1	85.0
P2	79
P3	83.8

The following are also known:

- Among the throws in the second round, only the last two were valid. Both throws enabled these players to qualify for the second phase, with one of them qualifying with the least score. None of these players won any medal.
- If a player throws first in a round AND he was also the last (among the current round) to throw in the previous round, then the players get a double. Two players got a double.

VIDEO SOLUTION

36. What was the final score (in m) of the silver-medalist?

- A 89.6
- B 88.4
- C 88.6
- D 87.2

Order

1. P1
2. P2
3. P3
4. P4
5. P5
6. P6
7. P7
8. P8
9. P9
10. P10

qualify for the second phase. In each of the rounds in the second phase, the players throw in decreasing order of their latest rank i.e. the player ranked 6 at that point throws first, followed by the player ranked 5 at that point and so on. The players ranked 1, 2, and 3 at the end of the sixth round receive gold, silver, and bronze medals respectively.

All the valid throws of the event were of distinct distances (as per stated measurement accuracy). The tables below show distances (in meters) covered by all valid throws in the first and the third round in the event.

Distances covered by all valid throws in the first round

Player	Distance (m)
P1	82.9
P2	85.1
P3	80.4
P4	82.7
P5	81.2
P6	81.1

Distances covered by all valid throws in the third round

Player	Distance (m)
P1	85.0
P2	80
P3	85.0

The following are also known:

- Among the throws in the second round, only the last two were valid. Both throws enabled these players to qualify for the second phase, with one of them qualifying with the least score. None of these players won any medal.
- If a player throws first in a round AND he was also the last (among the current round to throw in the previous round, then the player gets a double. Two players got a double.

VIDEO SOLUTION

37. Which of the following can be the final score (in m) of P8?

- A 81.9
- B 0
- C 82.7
- D 85.1

Order

1. P1
2. P2
3. P3
4. P4
5. P5
6. P6
7. P7
8. P8
9. P9
10. P10

qualify for the second phase. In each of the rounds in the second phase, the players throw in decreasing order of their latest rank i.e. the player ranked 6 at that point throws first, followed by the player ranked 5 at that point and so on. The players ranked 1, 2, and 3 at the end of the sixth round receive gold, silver, and bronze medals respectively.

All the valid throws of the event were of distinct distances (as per stated measurement accuracy). The tables below show distances (in meters) covered by all valid throws in the first and the third round in the event.

Distances covered by all valid throws in the first round

Player	Distance (m)
P1	82.9
P2	85.1
P3	80.4
P4	82.7
P5	81.2
P6	81.1

Distances covered by all valid throws in the third round

Player	Distance (m)
P1	85.0
P2	80
P3	85.0

The following are also known:

- Among the throws in the second round, only the last two were valid. Both throws enabled these players to qualify for the second phase, with one of them qualifying with the least score. None of these players won any medal.
- If a player throws first in a round AND he was also the last (among the current round to throw in the previous round, then the player gets a double. Two players got a double.

VIDEO SOLUTION



38. By how much did the gold medalist improve his score (in m) in the second phase?

- A 1.0
- B 2.0
- C 2.4
- D 1.2

Order

1. P1
2. P2
3. P3
4. P4
5. P5
6. P6
7. P7
8. P8
9. P9
10. P10

qualify for the second phase. In each of the rounds in the second phase, the players throw in decreasing order of their latest rank i.e. the player ranked 6 at that point throws first, followed by the player ranked 5 at that point and so on. The players ranked 1, 2, and 3 at the end of the sixth round receive gold, silver, and bronze medals respectively.

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Distances covered by all valid throws in the third round

Player	Distance (m)
P1	85.0
P2	80
P3	85.0

The following are also known:

- Among the throws in the second round, only the last two were valid. Both throws enabled these players to qualify for the second phase, with one of them qualifying with the least score. None of these players won any medal.
- If a player throws first in a round AND he was also the last (among the current round to throw in the previous round, then the player gets a double. Two players got a double.

VIDEO SOLUTION

Instructions [39 - 44]

Three reviewers Amal, Bimal, and Komal are tasked with selecting questions from a pool of 13 questions (Q01 to Q13). Questions can be created by external "subject matter experts" (SMEs) or by one of the three reviewers. Each of the reviewers either approves or disapproves a question that is shown to them. Their decisions lead to eventual acceptance or rejection of the question in the manner described below.

If a question is created by an SME, it is reviewed first by Amal, and then by Bimal. If both of them approve the question, then the question is accepted and is not reviewed by Komal. If both disapprove the question, it is rejected and is not reviewed by Komal. If one of them approves the question and the other disapproves it, then the question is reviewed by Komal. Then the question is accepted only if she approves it.

A question created by one of the reviewers is decided upon by the other two. If a question is created by Amal, then it is first reviewed by Bimal. If Bimal approves the question, then it is accepted. Otherwise, it is reviewed by Komal. The question is then accepted only if Komal approves it. A similar process is followed for questions created by Bimal, whose questions are first reviewed by Komal, and then by Amal only if Komal disapproves it. Questions created by Komal are first reviewed by Amal, and then, if required, by Bimal.

The following facts are known about the review process after its completion.

1. Q02, Q06, Q09, Q11, and Q12 were rejected and the other questions were accepted.
2. Amal reviewed only Q02, Q03, Q04, Q06, Q08, Q10, Q11, and Q13.
3. Bimal reviewed only Q02, Q04, Q06 through Q09, Q12, and Q13.
4. Komal reviewed only Q01 through Q05, Q07, Q08, Q09, Q11, and Q12.

39. How many questions were DEFINITELY created by Amal?

Right / Wrong by Komal. If both disapprove the question, it is rejected and is not reviewed by Komal. If one of them approves the question and the other disapproves it, then the question is reviewed by Komal. Then the question is accepted only if she approves it.

A question created by one of the reviewers is decided upon by the other two. If a question created by Amal, then it is first reviewed by Bimal. If Bimal approves the question, then it is accepted. Otherwise, it is reviewed by Komal. The question is then accepted only if Komal approves it. A similar process is followed for questions created by Bimal, whose questions are first reviewed by Komal, and then by Amal only if Komal disapproves it. Questions created by Komal are first reviewed by Amal, and then, if required, by Bimal.

The following facts are known about the review process after its completion.

[VIDEO SOLUTION](#)

40. How many questions were DEFINITELY created by Komal?

Right / Wrong by Komal. If both disapprove the question, it is rejected and is not reviewed by Komal. If one of them approves the question and the other disapproves it, then the question is reviewed by Komal. Then the question is accepted only if she approves it.

A question created by one of the reviewers is decided upon by the other two. If a question created by Amal, then it is first reviewed by Bimal. If Bimal approves the question, then it is accepted. Otherwise, it is reviewed by Komal. The question is then accepted only if Komal approves it. A similar process is followed for questions created by Bimal, whose questions are first reviewed by Komal, and then by Amal only if Komal disapproves it. Questions created by Komal are first reviewed by Amal, and then, if required, by Bimal.

The following facts are known about the review process after its completion.

[VIDEO SOLUTION](#)

100+ CAT Quant Questions



41. How many questions were DEFINITELY created by the SMEs?

Right / Wrong by Komal. If both disapprove the question, it is rejected and is not reviewed by Komal. If one of them approves the question and the other disapproves it, then the question is reviewed by Komal. Then the question is accepted only if she approves it.

A question created by one of the reviewers is decided upon by the other two. If a question created by Amal, then it is first reviewed by Bimal. If Bimal approves the question, then it is accepted. Otherwise, it is reviewed by Komal. The question is then accepted only if Komal approves it. A similar process is followed for questions created by Bimal, whose questions are first reviewed by Komal, and then by Amal only if Komal disapproves it. Questions created by Komal are first reviewed by Amal, and then, if required, by Bimal.

The following facts are known about the review process after its completion.

[VIDEO SOLUTION](#)

42. How many questions were DEFINITELY disapproved by Bimal?

- A 3
- B 4

D 5

 VIDEO SOLUTION

A either 0.25 or 0.75

B 0.25

C lies between 0.25 and 0.50

D lies between 0.25 and 0.75

VIDEO SOLUTION

cracku

A	7
B	4
C	5
D	2

 VIDEO SOLUTION

cracku

Quant

45. A shop owner bought a total of 64 shirts from a wholesale market that came in two sizes, small and large. The price of a small shirt was INR 50 less than that of a large shirt. She paid a total of INR 5000 for the large shirts, and a total of INR 1800 for the small shirts. Then, the price of a large shirt and a small shirt together, in INR, is

- A 175
- B 150
- C 200
- D 225

cracku

A shop owner bought a total of 64 shirts from a wholesale market that came in two sizes, small and large. The price of a small shirt was INR 50 less than that of a large shirt. She paid a total of INR 5000 for the large shirts, and a total of INR 1800 for the small shirts. Then, the price of a large shirt and a small shirt together, in INR, is

A) 175
B) 150
C) 200

$S + L = 64$
 $P \times S = 1800$
 $(P + 50) \times (64 - S) = 5000$
 $(100 + 50)(64 - S) = 5000$
 $(150)(64 - S) = 5000$
 $(64 - S) = \frac{5000}{150} = \frac{1000}{3}$
 $64 - S = \frac{1000}{3}$
 $S = 64 - \frac{1000}{3} = \frac{192 - 1000}{3} = \frac{-808}{3}$

VIDEO SOLUTION

CAT Expected Questions PDF



46. One day, Rahul started a work at 9 AM and Gautam joined him two hours later. They then worked together and completed the work at 5 PM the same day. If both had started at 9 AM and worked together, the work would have been completed 30 minutes earlier. Working alone, the time Rahul would have taken, in hours, to complete the work is

- A 11.5
- B 10
- C 12.5
- D 12

cracku

One day, Rahul started a work at 9 AM and Gautam joined him two hours later. They then worked together and completed the work at 5 PM the same day. If both had started at 9 AM and worked together, the work would have been completed 30 minutes earlier. Working alone, the time Rahul would have taken, in hours, to complete the work is

A) 11.5
B) 10
C) 12.5

$8R + 6G = W$
 $R \rightarrow 9am$
 $G \rightarrow 11am$
 $(9am)$

VIDEO SOLUTION

47. In a tournament, a team has played 40 matches so far and won 30% of them. If they win 60% of the remaining matches, their overall win percentage will be 50%. Suppose they win 90% of the remaining matches, then the total number of matches won by the team in the tournament will be

- A 80
- B 78
- C 84
- D 86

In a tournament, a team has played 40 matches so far and won 30% of them. If they win 60% of the remaining matches, their overall win percentage will be 50%. Suppose they win 90% of the remaining matches, then the total number of matches won by the team in the tournament will be _____.

A) 80
B) 78
C) 84
D) 82

Handwritten notes: 12 , 0.6 , $12 + 0.6N$, $40 + N$

VIDEO SOLUTION

48. The number of distinct pairs of integers (m, n) , satisfying $|1 + mn| < |m + n| < 5$ is:

The number of distinct pairs of integers (m, n) , satisfying $|1 + mn| < |m + n| < 5$ is:

Answer: _____

Handwritten notes: $(1+mn)^2 < (m+n)^2 < 25$, $1+m^2n^2+2mn < m^2+n^2+2mn$, $0 < m$

VIDEO SOLUTION

IMPORTANT CAT Important Topics

49. For a real number a , if $\frac{\log_{15} a + \log_{32} a}{(\log_{15} a)(\log_{32} a)} = 4$ then a must lie in the range

- A $2 < a < 3$
- B $3 < a < 4$
- C $4 < a < 5$
- D $a > 5$

For a real number a , if $\frac{\log_{15} a + \log_{32} a}{(\log_{15} a)(\log_{32} a)} = 4$ then a must lie in the range

A) $2 < a < 3$
B) $3 < a < 4$
C) $4 < a < 5$
D) $a > 5$

Handwritten notes: $\log_{15} a + \log_{32} a$, $\log_{15} a \times \log_{32} a$, $\frac{\log_{15} a}{\log 15} + \frac{\log_{32} a}{\log 32}$, $4 \frac{\log_{15} a}{\log 15} \times \frac{\log_{32} a}{\log 32}$

VIDEO SOLUTION

50. The total of male and female populations in a city increased by 25% from 1970 to 1980. During the same period, the male population increased by 40% while the female population increased by 20%. From 1980 to 1990, the female population increased by 25%. In 1990, if the female population is twice the male population, then the percentage increase in the total of male and female populations in the city from 1970 to 1990 is

- A 68.25
- B 68.75
- C 68.50
- D 69.25

The total of male and female populations in a city increased by 25% from 1970 to 1980. During the same period, the male population increased by 20% while the female population increased by 25%. From 1980 to 1990, the female population increased by 25%. In 1990, if the female population is twice the male population, then the percentage increase in the total of male and female populations in the city from 1970 to 1990 is

A) 68.25
B) 68.50
C) 68.75
D) 68.75

Handwritten solution:

$$\begin{array}{l} \text{Total} \quad X \\ \text{Males} \quad M \\ \text{Females} \quad F \end{array}$$

$$\begin{array}{l} 1970 \quad 1980 \quad 1990 \\ X \quad 1.25X \\ M \quad 1.2M \\ F \quad 1.25F \end{array}$$

$$(M+F) \times 1.25$$

$$(1.2M + 1.25F) \times 1.25$$

$$5M + 5F$$

VIDEO SOLUTION

51. Consider a sequence of real numbers, $x_1, x_2, x_3, \dots$ such that $x_{n+1} = x_n + n - 1$ for all $n \geq 1$. If $x_1 = -1$ then x_{100} is equal to

- A) 4849
B) 4949
C) 4950
D) 4850

Consider a sequence of real numbers, $x_1, x_2, x_3, \dots$ such that $x_{n+1} = x_n + n - 1$ for all $n \geq 1$. If $x_1 = -1$ then x_{100} is equal to

A) 4849
B) 4949
C) 4950
D) 4850

Handwritten solution:

$$\begin{array}{l} x_1 = -1 \\ x_2 = x_1 + 0 \\ x_3 = x_2 + 1 \\ x_4 = x_3 + 2 \\ \vdots \\ x_{100} = x_{99} + 99 \end{array}$$

$$x_1 + x_2 + \dots + x_{100} = x_{100}$$

VIDEO SOLUTION

CAT WhatsApp Group

cracku

WhatsApp icon

52. The arithmetic mean of scores of 25 students in an examination is 50. Five of these students top the examination with the same score. If the scores of the other students are distinct integers with the lowest being 30, then the maximum possible score of the toppers is

The arithmetic mean of scores of 25 students in an examination is 50. Five of these students top the examination with the same score. If the scores of the other students are distinct integers with the lowest being 30, then the maximum possible score of the toppers is

Answer: _____

Handwritten solution:

$$\begin{array}{l} 25 \quad 50 \\ 30 \quad 31 \quad \dots \quad 49 \\ 25 + 31 \end{array}$$

$$1250$$

$$1250 - 5T$$

VIDEO SOLUTION

53. One part of a hostel's monthly expenses is fixed, and the other part is proportional to the number of its boarders. The hostel collects ₹ 1600 per month from each boarder. When the number of boarders is 50, the profit of the hostel is ₹ 200 per boarder, and when the number of boarders is 75, the profit of the hostel is ₹ 250 per boarder. When the number of boarders is 80, the total profit of the hostel, in INR, will be

- A) 20200
B) 20500
C) 20800
D) 20000

cracku

One part of a hostel's monthly expenses is fixed, and the other part is proportional to the number of its boarders. The hostel collects ₹ 1600 per month from each boarder. When the number of boarders is 50, the profit of the hostel is ₹ 200 per boarder, and when the number of boarders is 75, the profit of the hostel is ₹ 250 per boarder. When the number of boarders is 80, the total profit of the hostel, in INR, will be

A) 20200
B) 20500
C) 20800
D) 20000

Expense = $F + K \cdot N$
Revenue = $1600N$
Total Profit = $1600N - F - K \cdot N$
Profit per boarder = $\frac{1600N - F - K \cdot N}{N}$

When $N = 50$, Profit = 200
When $N = 75$, Profit = 250

When $N = 80$, Profit = ?

54. The cost of fencing a rectangular plot is ₹ 200 per ft along one side, and ₹ 100 per ft along the three other sides. If the area of the rectangular plot is 60000 sq. ft, then the lowest possible cost of fencing all four sides, in INR, is

A) 120000
B) 90000
C) 100000
D) 160000

VIDEO SOLUTION

54. The cost of fencing a rectangular plot is ₹ 200 per ft along one side, and ₹ 100 per ft along the three other sides. If the area of the rectangular plot is 60000 sq. ft, then the lowest possible cost of fencing all four sides, in INR, is

- A) 120000
B) 90000
C) 100000
D) 160000

cracku

The cost of fencing a rectangular plot is ₹ 200 per ft along one side, and ₹ 100 per ft along the three other sides. If the area of the rectangular plot is 60000 sq. ft, then the lowest possible cost of fencing all four sides, in INR, is

A) 120000
B) 90000
C) 100000
D) 160000

Cost = $200L + 100B$
Area = $L \times B = 60000$
Cost = $200L + 100B$
Cost = $200L + 100B$
Cost = $200L + 100B$

VIDEO SOLUTION

cracku

CAT Mock Test

Mocks designed exactly like the actual CAT

MOCK

55. A park is shaped like a rhombus and has area 96 sq m. If 40 m of fencing is needed to enclose the park, the cost, in INR, of laying electric wires along its two diagonals, at the rate of ₹125 per m, is

cracku

A park is shaped like a rhombus and has area 96 sq m. If 40 m of fencing is needed to enclose the park, the cost, in INR, of laying electric wires along its two diagonals, at the rate of ₹125 per m, is

Answer: _____

Area = $\frac{d_1 \times d_2}{2} = 96$
Perimeter = $4a = 40$
Side = $a = 10$
Diagonals = d_1, d_2

VIDEO SOLUTION

56. A tea shop offers tea in cups of three different sizes. The product of the prices, in INR, of three different sizes is equal to 800. The prices of the smallest size and the medium size are in the ratio 2 : 5. If the shop owner decides to increase the prices of the smallest and the medium ones by INR 6 keeping the price of the largest size unchanged, the product then changes to 3200. The sum of the original prices of three different sizes, in INR, is

A tea shop offers tea in cups of three different sizes. The product of the prices, in INR, of three different sizes is equal to 800. The prices of the smallest size and the medium size are in the ratio $\frac{2}{3}$. If the shop owner decides to increase the prices of the smallest and the medium ones by INR 4 keeping the price of the largest size unchanged, the product then changes to 3200. The sum of the original prices of three different sizes, in INR, is

Answer:

Small Medium Large
 $2x$ $3x$ L
 $(2+4)$ $(3+6)$ L
 $8x$ $18x$ L
 800
 3200
 $5x^2$

VIDEO SOLUTION

57. Mira and Amal walk along a circular track, starting from the same point at the same time. If they walk in the same direction, then in 45 minutes, Amal completes exactly 3 more rounds than Mira. If they walk in opposite directions, then they meet for the first time exactly after 3 minutes. The number of rounds Mira walks in one hour is

Mira and Amal walk along a circular track, starting from the same point at the same time. If they walk in the same direction, then in 45 minutes, Amal completes exactly 3 more rounds than Mira. If they walk in opposite directions, then they meet for the first time exactly after 3 minutes. The number of rounds Mira walks in one hour is

Answer:

Circumference = C
 Amal = M
 Mira = A
 $3M + 3A = C$
 $45A = C$

VIDEO SOLUTION



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58. If a certain weight of an alloy of silver and copper is mixed with 3 kg of pure silver, the resulting alloy will have 90% silver by weight. If the same weight of the initial alloy is mixed with 2 kg of another alloy which has 90% silver by weight, the resulting alloy will have 84% silver by weight. Then, the weight of the initial alloy, in kg, is

- A 3.5
- B 2.5
- C 3
- D 4

If a certain weight of an alloy of silver and copper is mixed with 3 kg of pure silver, the resulting alloy will have 90% silver by weight. If the same weight of the initial alloy is mixed with 2 kg of another alloy which has 90% silver by weight, the resulting alloy will have 84% silver by weight. Then, the weight of the initial alloy, in kg, is

Answer:

A) 3.5 B) 2.5 C) 3 D) 4

$3+S$
 90%
 $2+S$
 90%
 84%

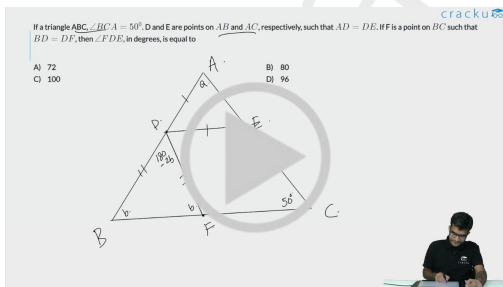
VIDEO SOLUTION

59. If a triangle ABC, $\angle BCA = 50^\circ$. D and E are points on AB and AC, respectively, such that $AD = DE$. If F is a point on BC such that $BD = DF$, then $\angle FDE$, in degrees, is equal to

- A 72
- B 80

C 100

D 96



VIDEO SOLUTION

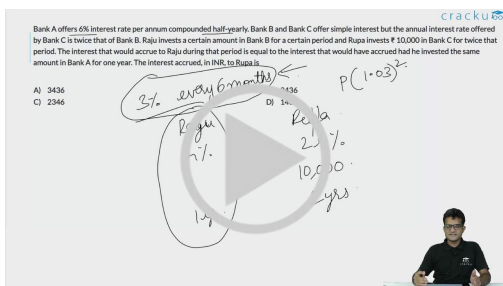
60. Bank A offers 6% interest rate per annum compounded half-yearly. Bank B and Bank C offer simple interest but the annual interest rate offered by Bank C is twice that of Bank B. Raju invests a certain amount in Bank B for a certain period and Rupa invests ₹ 10,000 in Bank C for twice that period. The interest that would accrue to Raju during that period is equal to the interest that would have accrued had he invested the same amount in Bank A for one year. The interest accrued, in INR, to Rupa is

A 3436

B 2436

C 2346

D 1436



VIDEO SOLUTION



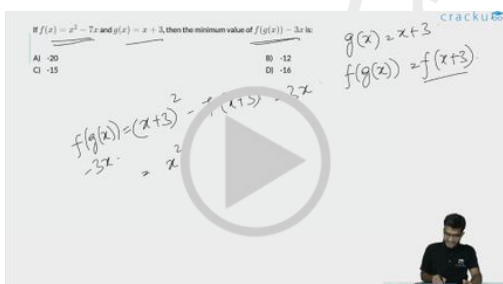
61. If $f(x) = x^2 - 7x$ and $g(x) = x + 3$, then the minimum value of $f(g(x)) - 3x$ is:

A -20

B -12

C -15

D -16



VIDEO SOLUTION

62. Anil can paint a house in 12 days while Barun can paint it in 16 days. Anil, Barun, and Chandu undertake to paint the house for ₹ 24000 and the three of them together complete the painting in 6 days. If Chandu is paid in proportion to the work done by him, then the amount in INR received by him is

Aril can paint a house in 12 days while Barun can paint it in 16 days. Aril, Barun, and Chandu undertake to paint the house for ₹ 24000 and the three of them together complete the painting in 6 days. If Chandu is paid in proportion to the work done by him, then the amount in INR received by him is _____

Answer: _____

Handwritten solution:
 4 units/day A = 12 days
 2 → 16 days
 A+ → 6 days
 Total = LCM(12, 16) = 48 units

VIDEO SOLUTION

63. If n is a positive integer such that $(\sqrt[n]{10})(\sqrt[n]{10})^2 \dots (\sqrt[n]{10})^n > 999$, then the smallest value of n is _____

If n is a positive integer such that $(\sqrt[n]{10})(\sqrt[n]{10})^2 \dots (\sqrt[n]{10})^n > 999$, then the smallest value of n is _____

Answer: _____

Handwritten solution:
 $10^{\frac{1}{n}} \times 10^{\frac{2}{n}} \times \dots \times 10^{\frac{n-1}{n}} \times 10^1 > 10^3$
 $10^{\frac{1+2+\dots+(n-1)}{n} + 1} > 10^3$
 $10^{\frac{n(n-1)}{2n} + 1} > 10^3$
 $10^{\frac{n-1}{2} + 1} > 10^3$
 $10^{\frac{n-1}{2} + 1} > 10^3$
 $\frac{n-1}{2} + 1 > 3$
 $\frac{n-1}{2} > 2$
 $n-1 > 4$
 $n > 5$
 Smallest value of n is 6

VIDEO SOLUTION



CAT Syllabus PDF



64. A four-digit number is formed by using only the digits 1, 2 and 3 such that both 2 and 3 appear at least once. The number of all such four-digit numbers is _____

A four-digit number is formed by using only the digits 1, 2 and 3 such that both 2 and 3 appear at least once. The number of all such four-digit numbers is _____

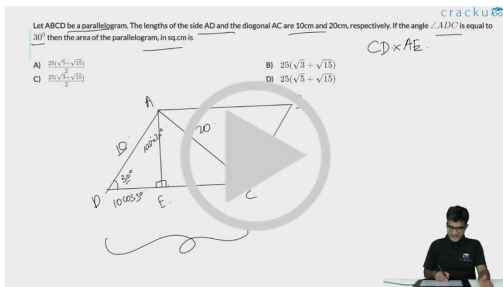
Answer: _____

Handwritten solution:
 Total 4-digit numbers using digits 1, 2, 3 = $3^4 = 81$
 Numbers with only 1s = 1 (1111)
 Numbers with only 2s = 1 (2222)
 Numbers with only 3s = 1 (3333)
 Numbers with only 1s and 2s = $2^4 - 2 = 14$
 Numbers with only 1s and 3s = $2^4 - 2 = 14$
 Numbers with only 2s and 3s = $2^4 - 2 = 14$
 Total numbers with at least one 2 and one 3 = $81 - 1 - 1 - 1 - 14 - 14 - 14 = 39$

VIDEO SOLUTION

65. Let ABCD be a parallelogram. The lengths of the side AD and the diagonal AC are 10cm and 20cm, respectively. If the angle $\angle ADC$ is equal to 30° then the area of the parallelogram, in sq.cm is _____

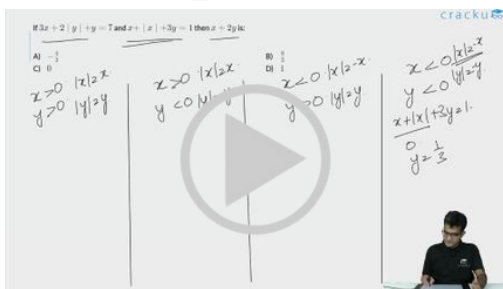
- A $\frac{25(\sqrt{5} + \sqrt{15})}{2}$
- B $25(\sqrt{3} + \sqrt{15})$
- C $\frac{25(\sqrt{3} + \sqrt{15})}{2}$
- D $25(\sqrt{5} + \sqrt{15})$



VIDEO SOLUTION

66. If $3x + 2|y| + y = 7$ and $x + |x| + 3y = 1$ then $x + 2y$ is:

- A $-\frac{4}{3}$
 B $\frac{8}{3}$
 C 0
 D 1



VIDEO SOLUTION



Answers

VARC

1.D	2.A	3.C	4.C	5.C	6.D	7.C	8.C
9.D	10.D	11.C	12.C	13.B	14.B	15.A	16.C
17.3412	18.2314	19.4	20.D	21.3	22.D	23.C	24.3214

LRDI

25.D	26.1	27.2	28.D	29.D	30.A	31.D	32.C
33.C	34.D	35.A	36.C	37.C	38.C	39.3	40.1
41.3	42.B	43.D	44.C				

Quant

45.C	46.B	47.C	48.12	49.C	50.B	51.D	52.92
53.B	54.A	55.3500	56.34	57.8	58.C	59.B	60.B
61.D	62.3000	63.6	64.50	65.B	66.C		

Explanations

VARC

1.D

The fundamental unity of humanity is the theme of Mr. Chomsky's universal grammar, and of this exciting book.

Throughout the passage, the author seems to support the points made by Mr Pinker. The above line also shows that the opinion of the author towards the book is positive, and the author does not think that the book is racist in any way, but promotes unity and cohesion. Option D captures this point correctly and is the answer.

So according to Mr. Pinker, the roots of language must be in the genes, but there cannot be a "grammar gene" any more than there can be a gene for the heart or any other complex body structure. This proposition will undoubtedly raise the hackles of some behavioral psychologists and anthropologists, for it apparently contradicts the liberal idea that human behavior may be changed for the better by improvements in culture and environment....

The book does not support that a complex anatomical structure like a 'voice box' plays a key role in determining language acquisition skills. Nor does it support the role of culture and environment in shaping human behaviour Options A and B are eliminated.

Option C portrays the book as racist, which is directly in contradiction with the author's opinion. C is eliminated too.

[VIDEO SOLUTION](#)

2.A

A half-century ago, this would have been pooh-poohed as a "black box" theory, since one could not actually pinpoint this grammatical faculty in a specific part of the brain, or describe its functioning. But now things are different. Neurosurgeons [have now found that this] "black box" is situated in and around Broca's area, on the left side of the forebrain. . . .

On the other hand, the "language instinct," when it first appeared among our most distant hominid ancestors, must have given them a selective reproductive advantage over their competitors (including the ancestral chimps). . . .

He effectively disposes of all claims that intelligent nonhuman primates like chimps have any abilities to learn and use language.

The above excerpts provide support for Options B, C, and D respectively. Option A is in direct contradiction with Option D, and hence, is the answer.

VIDEO SOLUTION

3. C

Unlike Mr. Chomsky, Mr. Pinker firmly places the wiring of the brain for language within the framework of Darwinian natural selection and evolution.

The passage suggests that Mr. Pinker and Mr. Chomsky agree on almost all topics. However, the above line indicates that they both disagreed on the application of the Darwinian framework to explain language instinct. Where Mr. Pinker was in favour of the same, Mr. Chomsky was against. Hence, Option C is the answer.

VIDEO SOLUTION

4. C

Since this message was couched in terms of Chomskyan theoretical linguistics, in discourse so opaque that it was nearly incomprehensible even to some scholars, many people did not hear it. Now, in a brilliant, witty and altogether satisfying book, Mr. Chomsky's colleague Steven Pinker . . . has brought Mr. Chomsky's findings to everyman.

From the above excerpt, it is clear that Mr. Pinker's style of writing is much more comprehensible to the common man. Hence, their writing styles are quite different. Also, the above excerpt mentions that the book brings Mr. Chomsky's findings to everyman, hence, it is clear that it draws heavily from the findings. Options A and D are eliminated.

Unlike Mr. Chomsky, Mr. Pinker firmly places the wiring of the brain for language within the framework of Darwinian natural selection and evolution.

The above excerpt shows that they both disagreed on a certain point. Hence, Option B is eliminated too.

Option C finds no mention in the passage, hence, is the answer.

VIDEO SOLUTION

5. C

The passage starts by highlighting that the term 'unconscious', widely held today, came in conception not long ago. With the coining of this term, many unrelated activities/ideas found a common umbrella under which they could be categorized and also allowed them to prosper. The author then writes the following line, which gives us a clear conception of the main theme:

Thus, the massive introduction of the term unconscious by Hartmann in 1869 appeared to focalize many stray thoughts, affording a temporary feeling that a crucial step had been taken forward, a comprehensive knowledge gained a knowledge that required only further elaboration, explication, and unfolding in order to bring in a bounty of higher understanding.

Thus, the passage is about the assembly of many stray thoughts under the banner of the unconscious. Option C perfectly captures this, and hence, is the answer.

The author does not primarily deal with the unconscious as a part of the mind. Nor does he focus upon the expansion of the vocabulary of the mind and the soul. Thus, Options A and B can be rejected.

'Psychical research' is not the main focus of the passage. The author says that the term allowed certain 'psychic' activities to flourish. He does not focus on the term as an object of psychical research. Hence, Option D can be eliminated too.

VIDEO SOLUTION



6. D

Let us try to break the sentence down and interpret its meaning:

"The enrichments of literary and intellectual language led to an altered understanding of the meanings that underlie time-honored expressions and traditional catchwords."

In simple words [Enrichments of language] led to [change in understanding] of [time-honoured expressions].

In the context of the passage, the line means that when the terms related to 'the unconscious' were coined, they enriched the vocabulary of the language and this, in turn, changes the meanings of many old expressions related to this term.

Option D comes the closest in capturing the meaning, and hence, is the answer.

B: The meanings of the catchwords were altered. They were not enriched. Can be eliminated.

C: The catchwords did not cause a change. Their own meaning was changed. Can be eliminated.

VIDEO SOLUTION

7. C

Unconscious is the primary focus of the passage. Since D does not have that as a main point, it can be eliminated.

Dreams find a single, small mention as an example in the passage. Hence, Option A can be eliminated too.

Insanity finds a small mention in the passage and is not a main point. Hence, Option B is incorrect.

The author initially deals with how the enrichment of vocabulary on the matter of unconscious has a deep effect and how this later became a subject of psychoanalysis. Hence, Option C is the correct answer.

VIDEO SOLUTION

8. C

The "unconscious" burst the shell of conventional language, coined as it had been to embody the fleeting ideas and the shifting conceptions of several generations until, finally, it became fixed and defined in specialized terms within the realm of medical psychology and Freudian psychoanalysis.

In the passage, the author has clearly outlined the importance of linguistic developments in helping the knowledge of the field grow. Since the option is not extreme in certainty ('may' not have happened), Option A can be inferred.

Significantly, most if not all of these practices (for example, hypnotism in surgery or psychological magnetism) originated in the waning years of the eighteenth century and during the early decades of the nineteenth century, as did some of the disciplines (such as psychology and psychical research). The majority of topics too were either new or assumed hitherto unknown colors. Thus, before 1790, few if any spoke, in medical terms, of the affinity between creative genius and the hallucinations of the insane . . .

From the above excerpt, we can infer that the affinity between genius and insanity was not looked into before the 18th century.

At the same time, once coined, powerful new ideas attracted to themselves a whole host of seemingly unrelated issues, practices, and experiences, creating a peculiar network of preoccupations that as a group had not existed before.

The above excerpt and the examples the author provides after this excerpt can help us infer that as the knowledge of the mind grew, unrelated activities found a common title. Option D can be inferred.

The passage does not imply anywhere that the new conceptions were able to provide new knowledge only because some fields were established. Option C is out of the scope of the passage and cannot be inferred.

VIDEO SOLUTION

9. D

Back in the early 2000s, an awesome thing happened in the New X-Men comics. Our mutant heroes had been battling giant robots called Sentinels for years, but suddenly these mechanical overlords spawned a new threat: Nano-Sentinels! Not content to rule Earth with their metal fists, these tiny robots invaded our bodies at the microscopic level. Infected humans were slowly converted into machines, cell by cell.

The first paragraph talks about the X-men comics, in which the mutant heroes, that X-Men, has been battling giant robots called sentinels. But these Sentinels then developed Nano-Sentinels, which could invade bodies at the microscopic level, and the heroes would now have to fight them too. Option D perfectly captures this, and hence, is the answer.

Option A is incorrect. X-men were battling the Sentinels before the invention of Nano-Sentinels. Hence, the origin of X-men is different.

Option B is incorrect. The mechanical overlords made Nano-Sentinels to convert people into machines. It has not been said that the people were converted into the mechanical lords themselves.

VIDEO SOLUTION

10. D

We will check which option when proven false will support the passage:

A: Robots becoming a part of everyday life is neither supported nor opposed in the passage. Thus, Option A is not the answer.

B: *Instead of gears and wires, it's assembled from two kinds of molecules - some that love water and some that avoid it - which interact to allow the bot to skate on top of a pond.*

Option B has been clearly mentioned in the passage. Hence, if it is proven false, it will contradict the passage. Option B is not the answer.

C: *One team of researchers, for example, has developed a different kind of hydrogel that becomes sticky when exposed to a low-voltage zap of electricity and then stops being sticky when the electricity is switched off.*

Option C has been mentioned in the passage. Hence, if it is proven false, it will contradict the passage. Option C is not the answer.

D: *No, they won't be invading our bodies and turning us into Sentinels...*

Option D is just the opposite of what has been given in the passage. Hence, if Option D is false, it would support the passage.

VIDEO SOLUTION

CAT Percentile Predictor



11. C

A: Sentinel robots are just fiction that is mentioned in the passage to introduce the new wave of development that has taken place. Option A is eliminated.

B: The author has introduced X-men as an example only. His arguments are not related to the creation of X-men in any way. Option B can be eliminated.

C: *.....but some of them could one day swim through our bloodstream to heal us.*

Throughout the passage, the author is trying to highlight the positives of the new robots. Hence, a direct extension of the argument would be the robots healing us at a microscopic level, as is hinted in the above excerpt. Option C is the answer.

D: Option D, though not entirely incorrect, is not a direct extension of the arguments presented in the passage. Unlike Option C, D has not been hinted at in the passage. Hence, it can be eliminated too.

VIDEO SOLUTION

12. C

The author first introduces an arc of a comic book where nano-robots are used. He then goes on to show how that fiction is increasingly becoming reality. He then goes on to describe the various features present in today's nano-robots. Option C comes the closest in capturing this point, and hence, is the answer.

Option A is just one of the features of the modern nano-robots and is not the focus of the passage.

Hydrophilic and Hydrophobic materials are not the main point of contention here. Option B can be eliminated.

The author uses the example of X-men to introduce the development of technology today. His main contention is not the content of the comic books and how it would be affected by recent developments in technology. Option D can be eliminated too.

[VIDEO SOLUTION](#)

13. B

A: We cannot infer that the 'arrow of time' has not been tested for atomic clocks. Option A can be eliminated.

B: It has been given in the Option that since quantum computers place more emphasis on their clock's accuracy, they would produce more heat.

The researchers found that as they increased the clock's accuracy, the heat produced in the system grew, increasing the entropy of its surroundings by jostling nearby particles...

The passage supports this inference. B is the answer.

C: *The relationship that the researchers found is a limit on the accuracy of a clock, so it doesn't mean that a clock that creates the most possible entropy would be maximally accurate - hence a large, inefficient grandfather clock isn't more precise than an atomic clock.*

The passage gives a specific example of an inefficient grandfather clock. We cannot infer whether all grandfather clocks are efficient or not.

D: *The clock consists of a 50-nanometre-thick membrane of silicon nitride, vibrated by an electric current.*

The clock uses electric current to produce vibrations and not the other way around. Option D can be eliminated.

[VIDEO SOLUTION](#)

14. B

There is an evident confusion between Option B and Option C; however, the official answer key marked Option B as the correct choice. Let us try to rationalise this decision. Options A and D can be understood from the passage:

Option A follows from {...*We can't be sure yet that these results are actually universal, though, because there are many types of clocks for which the relationship between accuracy and entropy haven't been tested...*}

Option D follows from {...*Understanding this relationship could be helpful for designing clocks in the future, particularly those used in quantum computers and other devices where both accuracy and temperature are crucial, says Ares...*}

Option C: Pay heed to the following excerpt from the passage - {...*The relationship that the researchers found is a limit on the accuracy of a clock, so it doesn't mean that a clock that creates the most possible entropy would be maximally accurate - hence a large, inefficient grandfather clock isn't more precise than an atomic clock. "It's a bit like fuel use in a car. Just because I'm using more fuel doesn't mean that I'm going faster or further," says Huber...*}

A simple correlation is being highlighted: higher accuracy means higher entropy; however, this does not necessarily imply that higher entropy translates to higher accuracy. The example of a grandfather clock is highlighted to emphasise this point: we will come across higher entropy in this case, but it does not mean that the grandfather clock is any more accurate than an atomic clock. In a way, the author tries to point out that the accuracy could very well be similar. This accuracy is not in absolute terms but in the way accuracy is defined by the author earlier in the passage. Thus, in a way, Option C matches the idea conveyed by the author

Option B: Pay heed to the following excerpt from the passage - {...*The researchers found that as they increased the clock's accuracy, the heat produced in the system grew, increasing the entropy of its surroundings by jostling nearby particles . . . "If a clock is more accurate, you are paying for it somehow," says Ares. In this case, you pay for it by pouring more ordered energy into the clock, which is then converted into entropy. "By measuring time, we are increasing the entropy of the universe," says Ares...*}

The discussion about the price paid appears to be distinct from the earlier segment wherein the author states that when we push for higher accuracy, we will come across more heat. While talking about the cost at which higher accuracy is achieved, the author states that we "pour in" more 'ordered energy' and this subsequently leads to higher entropy. Hence, the focus seems to be on the connection between accuracy and entropy than between heat and its role in creating higher accuracy. We cannot conclusively infer that the "ordered energy" stated in the latter half refers to the "heat" mentioned earlier on. Thus, claiming that heat is the price we pay for generating higher accuracy might be difficult to substantiate. Hence, Option B is distorted.

[VIDEO SOLUTION](#)

15. A

The relationship that the researchers found is a limit on the accuracy of a clock, so it doesn't mean that a clock that creates the most possible entropy would be maximally accurate - hence a large, inefficient grandfather clock isn't more precise than an atomic clock. "It's a bit like fuel use in a car. Just because I'm using more fuel doesn't mean that I'm going faster or further," says Hube

In the above excerpt, the author gives an example that though a large, inefficient grandfather clock would produce more entropy, it is not necessarily more precise than an atomic clock. Hence, if a clock produces more entropy, it does not mean that it would be more precise than a clock that produces less entropy. Then the mentioned statement is given as an example. If a car is going faster or further, it will definitely use more fuel. But if a car is using more fuel, then the converse is not true. It could just be possible that the mileage of the car is low. Option A comes the closest to capturing this idea, and hence, is the answer.

[VIDEO SOLUTION](#)



16. C

The maximum accuracy of a clock is directly related to how much disorder, or entropy, it creates every time it ticks.

The author highlights in the beginning of the passage that the accuracy associated with measuring time is directly related to how much entropy it creates while ticking. The author then goes on to talk about the relationship between accuracy and entropy, and how quantum mechanics and thermodynamics come in play here. Thus, the main keywords are the measurement of time, accuracy and entropy. Option C is the answer.

Electric current is just a small part of an example presented in the passage. Option A can be eliminated. The same is the case for Silicon Nitride and Membrane. These are just keywords associated with a particular experiment/example presented in the passage and are not important for the passage as a whole.

VIDEO SOLUTION

17. 3412

A quick read of the sentences tells us that the paragraph is about the unemployment caused by automation. The passage is best opened by 3, which provides the current state of unemployment. The politicians view globalisation as the factor exacerbating unemployment. 4 contrasts this by saying that expert analysis tells a different story. It is automation that could prove to be a crucial factor. 12 forms a pair, that further elucidate the kind and scope of impact that automation has on jobs. Hence, the correct sequence would be 3412.

VIDEO SOLUTION

18. 2314

A brief reading of the sentences tells us that the paragraph is about the different conceptions of relationships. 2 explains that communication is an important aspect here, and should be studied properly. 3 mentions a Foucauldian approach, that goes beyond this, and includes macrolevel regimes of truth. 1 then explains why the concept of regimes of truth is relevant here. 4 then aptly concludes the paragraph, implying how the micropractices within the relationships allude the importance of knowledge/power. Thus, the correct sequence would be 2314.

VIDEO SOLUTION

19. 4

A brief reading of the sentences suggests that the paragraph must be about the disparity in the representation of different genders. Sentences 1,2,3, and 5 are concerned with the problems that arise when the representation of females is less.

4, however, runs tangent to the discussion at hand. It talks about 'structural inequalities'. This sentence, if included in the paragraph, would render it incomplete as all the other sentences talk about gender inequality and not structural inequality. Thus, 4 is out of context here.

VIDEO SOLUTION

20. D

The main points of the passage are:

1. People increasingly view idleness as sin and industriousness as a virtue, pushing them into meaningless jobs.
2. On the other hand, this has also saved us from many of life's drudgeries.

A: Misses out on point 1.

B: A distortion. The author does not advocate idleness. Also, 1 is not covered properly.

C: Incorrect. 'led to greater idleness' is not implied anywhere in the passage.

D: Covers both the points aptly and is the answer.

VIDEO SOLUTION



21. 3

A brief reading of the sentences tells us that the paragraph is about Chinese business schools and how they stand in comparison to their western counterparts. 2 mentions that though they have a similar outlook, Chinese business schools have a different curriculum and are also different in what they have to offer. 1, 4, and 5 further talk about the peculiarity of the Chinese schools.

3, however, runs tangent to the discussion. It shifts the focus from Chinese schools and describes western schools. Hence, 3 is out of the context here.

VIDEO SOLUTION

22. D

The main points of the paragraph are as follows:

1. Intuition draws from a vast array of memories that our brain keeps in store.
2. When our brain recognises a pattern from past memories, neuron firing starts, which gives us the gut feeling of intuition.

A: Distortion: The passage does not give any detail about big data being based on rational thought.

B: Out of scope. The paragraph does not allude to whether the decisions based on intuition are better or worse.

C: Incorrect: The passage says that intuitive decisions are based on data.

D: Correctly covers the mentioned points and hence, is the answer.

VIDEO SOLUTION

23. C

The main points of the paragraph are:

1. The Brazilian economy has been stagnant, but the popular perception is that the times have improved.
2. The reasons are falling inequality and other important services.
3. Despite this, the economy is brutally unequal.

A: Ignorance on the part of Brazilians is not implied. What the author is saying that though things have improved for the ones who say so, others are still being dealt a rough hand.

B: The paragraph does not imply that the good economic indicators are being used as subterfuge to cover up the prevailing inequality.

C: Comes the closest in capturing the three points. Hence, is the answer.

D: It has a problem similar to Option A. Things have improved for that section of people. They are not ignorant, nor are they being misled into believing something.

[VIDEO SOLUTION](#)

24. 3214

A brief reading of the sentences suggests that the paragraph is about restitution of articles that were taken from colonies. 3 introduces the topic at hand: European museums are trying to come to terms with their colonial legacy by returning artefacts. 2 follows 3, providing the reason why this development has taken place. 1 then further describes the current situation: the legal obstacles this action could face. 4 concludes the paragraph by saying that other than these legal obstacles, institutions in France and Germany are seriously backing the move. Hence, the correct arrangement is 3214.

[VIDEO SOLUTION](#)

LRDI

25. D

Given that each of the bottles contains a volume of 50 ml each.

If 5 ml from bottle A which contains only P is mixed with 5 ml of bottle B and in the resultant mixture the presence of I was detected.

In order to detect the presence of I in this, there must be at least 10% impurity in the 10 ml which is equivalent to 1 ml. This must be from bottle B.

Hence 5 ml of solution from B must contain at least 1ml of impurity and since bottle B is of a total volume of 50 ml. This must contain at least 10 ml of impurity.

[VIDEO SOLUTION](#)



26. 1

The bottles contain either P(pure) or I(impure). The possible cases here are :

1- (P, P, P, P), 2-(P,P,P,I), 3-(P,P,I,I), 4-(P,I,I,I), 5-(I,I,I,I).

In the first case if all the four solutions are pure then taking equal volumes of all the four bottles will get the result to dispatch or not to dispatch.

In the second case if 3 bottles are pure and one impure taking equal volumes of all four bottles and testings will confirm the impurity and hence cannot be dispatched.

In the third case if 2 bottles are pure and two are impure taking equal volumes of all four bottles and testing will confirm the impurity and hence cannot be dispatched.

In the fourth case when only one bottle is pure taking equal volumes of all four bottles will confirm the impurity and hence cannot be dispatched.

In the fifth case if all four bottles are impure taking equal volumes of the four bottles will confirm the impurity and hence cannot be dispatched.

In all the cases a single test is enough to determine if the lot is to be dispatched or not.

[VIDEO SOLUTION](#)

27. 2

The percentage concentration of the impure solution is 80 percent.

When equal volumes of all four solutions are mixed.

Considering 10 ml of each we have impurity to be 2ml/40ml. The impurity concentration is less than 10 percent and hence cannot be recognized.

Similarly when equal volumes of one impure and 2 pure solutions are mixed.

The impurity in the solution is 2ml/30ml which is less than 10 percent and hence cannot be recognized.

Hence for detecting the impure solution we must use equal volumes of 2 solutions at a time.

Considering the three pure solutions to be P and the impure solution to be I.

P, P, P, I.

Considering equal volumes of solution from the bottle one bottle of P, and I. Testing this would recognize the impurity.

After this consider one bottle among the other 2 P bottles which are left and test this with one among the previously tested P, I.

If the one considered is I it will detect the impurity and confirms the bottle to be I.

If the one considered is P it will fail to detect the impurity and hence the other bottle will be I.

Hence a minimum of two tests are required to identify the bottle with the impurity.

[VIDEO SOLUTION](#)

28. D

The bottles could possibly be :

Case - 1 Pure, Impure, Impure, Impure.

Case-2, Pure , Pure, Impure, Impure.

Since the concentration in the impure bottle is 85 percent.

In case 1 when equal volumes from all the bottles are considered and mixed. The test result detects the impurity..

Since the overall concentration of impurity is greater than 10 percent.

Considering 10 ml from all four bottles.

The impure concentration is 4.5ml/40ml which is greater than 10. (15ml*3 = 4.5ml) (Impurity is detected)

For case 2 when all four bottles are considered. The case here has 2 pure and 2 impure bottles.

When equal volumes from all four bottles are mixed. The resultant concentration of impurity when 10 ml from each of the four solutions is considered :

The impure concentration is 3ml/40ml which is less than 10 percent.. (1.5ml*2 = 3ml). (Impurity is not detected.)

Hence in one possibility the impurity is detected and not detected in the other case. A single test is enough based on the results of which the number of pure and the number of impure bottles can be identified.

[VIDEO SOLUTION](#)

29. D

The total project month is the number of months Abani, Bahni, Danni, and Tinni individually worked for all the projects combined :

Abani - 2+2+5 = 9 months

Bahni - 2+4+3 = 9 months

Danni - 3+3+2+1 = 9 months

Tinni - 2+2+3+2 = 9 months.

The total employee month for all the five projects is the sum of the total employee-month is the sum of the number of months four employees worked in this project.

Project -1 = 2+2+2 = 6 months

Project -2 = 3+2 = 5 months

Project - 3 = 2+4+3 = 9 months.

Project - 4 = 5+2+3 = 10 months.

Project - 5 = 3+1+2 = 6 months.

Only statement 1 is true.

[VIDEO SOLUTION](#)

30. A

Abani, Banni, and Danni did not work on multiple projects simultaneously in a month

Tinni was the only person who worked on multiple projects which are project 4 and project 5 in the month of september.

[VIDEO SOLUTION](#)



31. D

Considering the information provided :

For project 1 : 3 months.

Project - 2: 3 months.

Project - 3: 5 months.

Project - 4: 5 months.

Project - 5: 4 months.

Among the given options option D is true which is project 3, project 4.

[VIDEO SOLUTION](#)

32. C

The annual completion index for different people is :

The weightage average of the completion percentage assigned from the different projects, with the weights being the corresponding number of months (s)he worked in these projects.

For Abani :

$$\frac{((100-2)+(100-2)+(80-5))}{2+2+5} = \frac{800}{9}$$

For Bahni :

$$\frac{((100-2)+(75-4)+(90-3))}{2+3+4} = \frac{770}{9}$$

For Danni :

$$\frac{((90-3)+(100-3)+(100-2)+(100-1))}{2+3+2+1} = \frac{870}{9}$$

For Tinni :

$$\frac{((80-2)+(100-2)+(100-3)+(100-2))}{2+2+3+2} = \frac{860}{9}$$

The descending order for the four people is :

Danni, Tinni, Abani, Bahni.

[VIDEO SOLUTION](#)

Explanation [33 - 38]:

Let us arrange the players in the order in which they throw in each round.

Round 1: Here the players throw in order of their initial seeds so the order is as follows:

Throw	Player
1	P1
2	P2
3	P3
4	P4
5	P5
6	P6
7	P7
8	P8
9	P9
10	P10

So, their rank at the start of Round 2 is in order of their throws in the first round. Also, we need to consider the same order for people having invalid throws.

Round 2: P2, P4, P8, P10 had the same relative rankings since they all have invalid throws, that is, 0 metres. Rest are arranged as per their throw distances.

Throw	Round 1	Round 2
1	P1	P7
2	P2	P5
3	P3	P9
4	P4	P1
5	P5	P6
6	P6	P3
7	P7	P2
8	P8	P4
9	P9	P8
10	P10	P10

Now, in round 2, only P8 and P10 had valid throws. Hence, their order will change at the start of Round 3, however, the remaining order stays the same. That is, P8 and P10 will move up in the table and occupy some higher places, whereas some of the others may move down consequently.

Round 3: In Round 3, we can see that P1 improved his score from 82.9 to 88.6. The other 2 participants did not improve their scores. Also after Round 3, P8 and P10 qualify, where one of P8 or P10 is at the sixth position. So at the end of Round 3, we can say that P6, P3, P2, and P4 are at the bottom 4 positions(ranks). One of P8 or P10 is at the sixth position. $P1 > P7 > P5 > P9$.

So at the end of round 3, the ranks are as follows:

P1
P7
P5
P9
P8/P10
P6
P3
P2
P4

The other person between P8/P10 can go anywhere between Rank 1 and Rank 5.

Now let us consider the two players who got a double. Doubles happen in the transition between rounds.

1 -> 2 - Not possible

2 -> 3 - Possible if P10 reaches Rank 1 after round 2.

3 -> 4 - P8/P10 who is the last among qualifying will be the first to throw. So, here it definitely happens.

4 -> 5 AND 5 -> 6 not possible.

So, after Round 2, definitely, P10 reaches the top of the ranking. P8 is at the bottom. Hence, after Round 3, P10 either retains rank 1 or P1 surpasses him and P10 becomes Rank 2.

So, two combinations are possible at the end of Round 3:

Rank after Round 3	
Case 1	Case 2
P10	P1
P1	P10
P7	P7
P5	P5
P9	P9
P8	P8

Now, we know that in each of the rounds in phase 2, only one player improves his score. Also, P8 and P10 cannot win medals. Hence, in case 1, three of P1, P7, P5 and P9 will improve their scores by x and reach the top 3 positions. However, the top 3 positions' distances are in AP.

$$P1 - 88.6 + x$$

$$P7 - 87.2 + x$$

$$P5 - 86.4 + x$$

$$P9 - 84.1 + x$$

The differences do not satisfy the condition. Hence, case 1 is invalidated.

Case 2: Here, P1 definitely wins a medal, and P10 does not. So, two of P7, P5 and P9 jumps above P10. Now, if we have three different people increasing their scores or distances in each of the three rounds, again we would not get a difference of 1 among the Gold, Silver and Bronze medallists. Hence, one of them increases his score twice and the other increases his score twice and none of them is P1.

Let us take the cases where P1 is individually the G, S and B medallists.

Case 1: P1 is a G medallist.

$$P1 - 88.6$$

The silver medallist is 87.6 and the bronze medallist is 86.6 metres. However, P10 has thrown for a distance that is greater than 87.2 metres. Hence, in this case, he would be the B medallist. Hence, this is not the right case.

Case 2: P1 is the S medallist.

P1 - 88.6

G - 89.6

B - 87.6

Now, if we see the differences

$89.6 - 87.2 = 2.4$

$87.6 - 86.4 = 1.2$

This satisfies the condition that P7 has increased his score twice to become the gold winner and P5 has increased it once to become the bronze winner.

Hence, P1 - Silver

P7 - Gold

P5 - Bronze

33. C

Hence, P8 and P10 got the doubles.

VIDEO SOLUTION

34. D

Hence, P1 won the silver medal.

VIDEO SOLUTION

35. A

P7 the gold winner had already received rank 1 at the end of Round 5. Hence, he was the last one to throw in the tournament.

VIDEO SOLUTION



36. C

P1 - 88.6 m.

VIDEO SOLUTION

37. C

P8 comes between P9 and P6.

Hence, $82.5 < P8 < 84.1$

$P8 = 82.7$

VIDEO SOLUTION

38. C

P7 improved by a total of 2.4.

VIDEO SOLUTION

Explanation [39 - 44]:

For a question created by externals(SME), the minimum number of reviews required is two and the maximum number of reviews is three to confirm the acceptance or the rejection.

For a question created by anyone among Amal, Bimal, Komal a question can be reviewed after one review or two reviews.

The information provided states :

Q02, Q06, Q09, Q11, and Q12 were rejected. Q01, Q03, Q04, Q05, Q07, Q08, Q10, Q13 were accepted.

For the questions :

Amal - Q02, Q03, Q04, Q06, Q08, Q10, Q11, and Q13.

Bimal - Q02, Q04, Q06, Q07, Q08, Q09, Q12, and Q13.

Komal - Q01, Q02, Q03, Q04, Q05, Q07, Q08, Q09, Q11, and Q12.

Q01 is reviewed by only Komal and is accepted hence Bimal must have prepared the question as it was accepted in a single review. (Bimal)

Q02-is reviewed by Amal, Bimal, and Komal and was rejected this has a possibility of being prepared by an external person and faced rejection by one among Amal and Bimal and Komal. (SME).

Q03-Was accepted and reviewed by Amal, Komal and hence must have been prepared by Bimal(Bimal).

Q04- Accepted and reviewed by the three of them and hence must have been prepared by an external expert. (SME).

Q05-Only reviewed by Komal and is accepted and hence must be prepared by Bimal. (Bimal)

Q06-Rejected and reviewed by Amal and Bimal. Hence could have been prepared by Komal or External and rejected by Bimal and Amal. (Komal/SME).

Q07-Accepted and reviewed by Bimal and Komal. Hence must have been prepared by Amal. (Amal).
 Q08- Reviewed by Amal, Bimal, and Komal and is accepted. Hence must have been prepared by SME. (SME)
 Q09- Reviewed by Bimal and Komal and is rejected. It must have been prepared by Amal and was rejected by both of them. (Amal).
 Q10- Was reviewed by Amal and was accepted. Must have been prepared by Komal. (Komal)
 Q11- Reviewed by Amal and Komal and was rejected. Must have been prepared by Bimal and rejected by both Amal and Komal. (Bimal)
 Q12- Reviewed by Bimal and Komal and was rejected. Must have been prepared by Amal and was rejected by both Bimal and Komal. (Amal).
 Q13- Reviewed by Amal and Bimal and was accepted. The question could have been prepared by Komal or SME. (Komal/SME).
39.3
 Amal definitely prepared questions Q07, Q09, Q12.

[VIDEO SOLUTION](#)

40.1
 Q10 was definitely prepared by Komal.

[VIDEO SOLUTION](#)



41.3
 Q02, Q04, Q08 were prepared SME

[VIDEO SOLUTION](#)

42.B
 Q06 irrespective of the question prepared by Komal/SME Bimal must reject the question.
 Q07, Q09, Q12 are prepared by Amal and were reviewed by Bimal first and Komal next for this to happen Bimal must reject these questions.
 Hence a total of four questions were rejected by Bimal.

[VIDEO SOLUTION](#)

43.D
 Amal reviewed the questions : Q02, Q03, Q04, Q06, Q08, Q10, Q11, and Q13.
 For Q2 - Amal can either approve or reject the question.
 For Q3- Amal must approve the question.
 For Q4 -Amal can either approve or reject the question.
 For Q06 - Amal must reject the question.
 For Q08 - Amal can either approve or reject the question.
 For Q10 - Amal must approve the question.
 For Q11 - Amal must reject the question.
 For Q13- Amal can either approve or reject the question.
 The approval ratio has a minimum of 2 questions only Q3, Q10, and a maximum of 6 questions Q02, Q3, Q04, Q08, Q10, Q13.
 Hence $2/8 = 0.25$ to $6/8 = 0.75$

[VIDEO SOLUTION](#)

44.C
 Amal could have possibly created the questions Q07, Q09, Q12 of which all three of them were reviewed by two people each, and hence they must have been rejected by the first reviewer.
 Bimal could have created the questions :
 Q01, Q03, Q05, Q 11.
 Q01, Q05 are reviewed only by Komal and hence did not have any rejection.
 Q03, Q11 faced at least one rejection.
 Hence a total of five questions.

[VIDEO SOLUTION](#)

Quant

45.C
 Let the number of large shirts be l and the number of small shirts be s.
 Let the price of a small shirt be x and that of a large shirt be x + 50.
 Now, $s + l = 64$
 $l(x+50) = 5000$

$$sx = 1800$$

Adding them, we get,

$$lx + sx + 50l = 6800$$

$$64x + 50l = 6800$$

Substituting $l = (6800 - 64x) / 50$, in the original equation, we get

$$\frac{(6800 - 64x)}{50} (x + 50) = 5000$$

$$(6800 - 64x)(x + 50) = 250000$$

$$6800x + 340000 - 64x^2 - 3200x = 250000$$

$$64x^2 - 3600x - 90000 = 0$$

$$\text{Solving, we get, } x = \frac{225 \pm 375}{8} = \frac{600}{8} \text{ or } -\frac{150}{8}$$

$$\text{SO, } x = 75$$

$$x + 50 = 125$$

$$\text{Answer} = 75 + 125 = 200.$$

Alternate approach: By options.

Hint: Each option gives the sum of the costs of one large and one small shirt. We know that large = small + 50

Hence, small + small + 50 = option.

$$\text{SMALL} = (\text{Option} - 50)/2$$

$$\text{LARGE} = \text{Small} + 50 = (\text{Option} + 50)/2$$

Option A and Option D gives us decimal values for SMALL and LARGE, hence we will consider them later.

Lets start with **Option B.**

$$\text{Large} = 150 + 50 / 2 = 100$$

$$\text{Small} = 150 - 50 / 2 = 50$$

$$\text{Now, total shirts} = 5000/100 + 1800/50 = 50 + 36 = 86 \text{ (X - This is wrong)}$$

Option C -

$$\text{Large} = 200 + 50 / 2 = 125$$

$$\text{Small} = 200 - 50 / 2 = 75$$

$$\text{Total shirts} = 5000/125 + 1800/75 = 40 + 24 = 64 \text{ (This is the right answer)}$$

[VIDEO SOLUTION](#)



46. **B**

Let Rahul work at a units/hr and Gautam at b units/hour

Now as per the condition :

$$8a + 6b = 7.5a + 7.5b$$

$$\text{so we get } 0.5a = 1.5b$$

$$\text{or } a = 3b$$

$$\text{Therefore total work} = 8a + 6b = 8a + 2a = 10a$$

$$\text{Now Rahul alone takes } 10a/10 = 10 \text{ hours.}$$

[VIDEO SOLUTION](#)

47. **C**

Initially number of matches = 40

Now matches won = 12

Now let remaining matches be x

Now number of matches won = $0.6x$

Now as per the condition :

$$\frac{(12 + 0.6x)}{40 + x} = \frac{1}{2}$$

$$24 + 1.2x = 40 + x$$

$$0.2x = 16$$

$$x = 80$$

$$\text{Now when they won } 90\% \text{ of remaining} = 80(0.9) = 72$$

$$\text{So total won} = 84$$

[VIDEO SOLUTION](#)

48. **12**

Let us break this up into 2 inequations [Let us assume x as m and y as n]

$$|1 + mn| < |m + n|$$

$$|m + n| < 5$$

Looking at these expressions, we can clearly tell that the graphs will be symmetrical about the origin.

Let us try out with the first quadrant and extend the results to the other quadrants.

We will also consider the +X and +Y axes along with the quadrant.

So, the first inequality becomes,

$$1 + mn < m + n$$

$$1 + mn - m - n < 0$$

$$1 - m + mn - n < 0$$

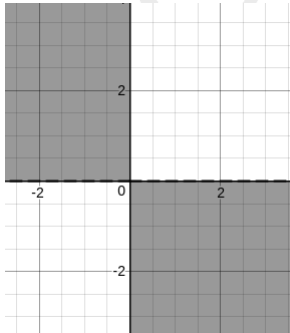
$$(1-m) + n(m-1) < 0$$

$$(1-m)(1-n) < 0$$

$$(m-1)(n-1) < 0$$

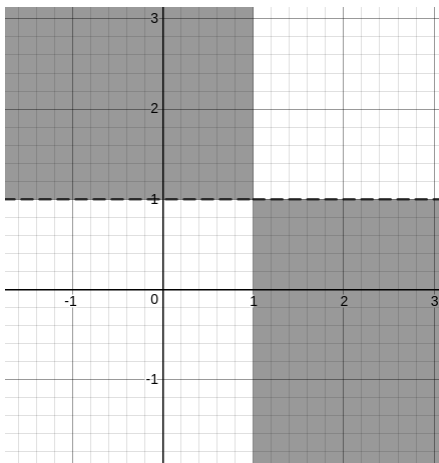
Let us try to plot the graph.

If we consider only $mn < 0$, then we get

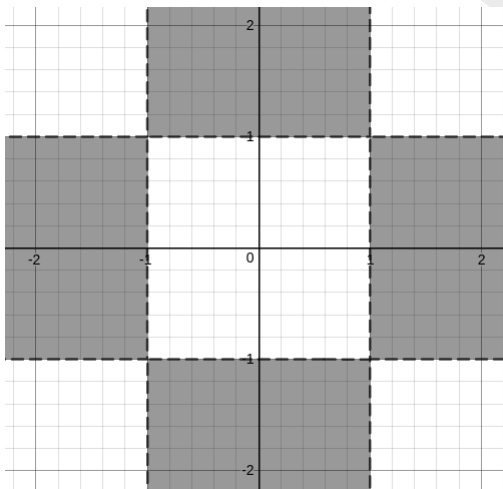


But, we have $(m-1)(n-1) < 0$, so we need to shift the graphs by one unit towards positive x and positive y.

So, we have,



But, we are only considering the first quadrant and the +X and +Y axes. Hence, if we extend, we get the following region.



So, if we look for only integer values, we get

$(0,2), (0,3), \dots$

$(0,-2), (0,-3), \dots$

$(2,0), (3,0), \dots$

$(-2,0), (-3,0), \dots$

Now, let us consider the other inequality as well, in which $|x + y| < 5$

Since one of the values is always zero, the modulus of the other value is less than or equal to 4.

Hence, we get

(0,2), (0,3), (0,4)

(0,-2), (0,-3), (0,-4)

(2,0), (3,0), (4,0)

(-2,0), (-3,0), (-4,0)

Hence, a total of 12 values.

[VIDEO SOLUTION](#)

49.C

We have : $\frac{\log_{15} a + \log_{32} a}{(\log_{15} a)(\log_{32} a)} = 4$

We get $\frac{\left(\frac{\log a}{\log 15} + \frac{\log a}{\log 32}\right)}{\frac{\log a}{\log 15} \times \frac{\log a}{\log 32}} = 4$

we get $\log a (\log 32 + \log 15) = 4 (\log a)^2$

we get $(\log 32 + \log 15) = 4 \log a$

$= \log 480 = \log a^4$

$= a^4 = 480$

so we can say a is between 4 and 5.

[VIDEO SOLUTION](#)

50.B

Let us solve this question by assuming values(multiples of 100) and not variables(x).

Since we know that the female population was twice the male population in 1990, let us assume their respective values as 200 and 100.

Note that while assuming numbers, some of the population values might come out as a fraction(which is not possible, since the population needs to be a natural number). However, this would not affect our answer, since the calculations are in ratios and percentages and not real values of the population in any given year.

	1970	1980	1990
Male			100
Female			200

Now, we know that the female population became 1.25 times itself in 1990 from what it was in 1980.

Hence, the female population in 1980 = $200/1.25 = 160$

Also, the female population became 1.2 times itself in 1990 from what it was in 1970.

Hence, the female population in 1970 = $160/1.2 = 1600/12 = 400/3$

	1970	1980	1990
Male			100
Female	400/3	160	200

Let the male population in 1970 be x. Hence, the male population in 1980 is 1.4x.

	1970	1980	1990
Male	x	1.4x	100
Female	400/3	160	200

Now, the total population in 1980 = 1.25 times the total population in 1970.

Hence, $1.25 (x + 400/3) = 1.4x + 160$

Hence, $x = 400/9$.

Population change = $300 - 400/9 - 400/3 = 300 - 1600/9 = 1100/9$

percentage change = $\frac{\frac{1100}{9}}{\frac{1600}{9}} \times 100 = \frac{1100}{16} \% = 68.75\%$

[VIDEO SOLUTION](#)

100+ CAT Quant Questions



51.D

Given $x_{n+1} = x_n + n - 1$ and $x_1 = -1$.

Considering

$x_1 = -1$. (1)

$x_2 = x_1 + 1 - 1 = x_1 + 0$ (2)

$x_3 = x_2 + 2 - 1 = x_2 + 1$ (3)

$x_4 = x_3 + 3 - 1 = x_3 + 2$ (4)

$x_{100} = x_{99} + 98$ (100)

Adding the LHS and RHS for the hundred equations we have:

$(x_1 + x_2 + \dots + x_{100}) = (-1 + 0 + \dots + 98) + (x_1 + x_2 + \dots + x_{99})$

Subtracting this we have :

$$(x1 + \dots + x100) - (x1 + \dots + x99) = \frac{(98-99)}{2} - 1.$$

$$x100 = 4851 - 1 = 4850$$

Alternatively

$$x_1 = -1$$

$$x_2 = x_1 + 1 - 1 = x_1 = -1$$

$$x_3 = x_2 + 2 - 1 = x_2 + 1 = -1 + 1 = 0$$

$$x_4 = x_3 + 3 - 1 = x_3 + 2 = 0 + 2 = 2$$

$$x_5 = x_4 + 4 - 1 = x_4 + 3 = 2 + 3 = 5$$

.....

If we observe the series, it is a series that has a difference between the consecutive terms in an AP.

Such series are represented as $t(n) = a + bn + cn^2$

We need to find $t(100)$.

$$t(1) = -1$$

$$a + b + c = -1$$

$$t(2) = -1$$

$$a + 2b + 4c = -1$$

$$t(3) = 0$$

$$a + 3b + 9c = 0$$

Solving we get,

$$b + 3c = 0$$

$$b + 5c = 1$$

$$c = 0.5$$

$$b = -1.5$$

$$a = 0$$

Now,

$$t(100) = (-1.5)100 + (0.5)100^2 = -150 + 5000 = 4850$$

[VIDEO SOLUTION](#)

52.92

Let sum of marks of students be x

Now therefore $x = 25 \times 50 = 1250$

Now to maximize the marks of the toppers

We will minimize the marks of 20 students

so their scores will be (30,31,32.....49)

let score of toppers be y

so we get $5y + \frac{20}{2}(79) = 1250$

we get $5y + 790 = 1250$

$$5y = 460$$

$$y = 92$$

So scores of toppers = 92

[VIDEO SOLUTION](#)

53.B

Profit per boarder = Total profit / Number of boarders.

Let the number of boarders be n.

$$\text{Profit/boarder} = 1600 - (\text{Total cost}/n)$$

Let the total cost be $a + bn$, where a = fixed, and b is the variable additional cost per boarder.

$$\text{Profit/boarder} = 1600 - (a + bn)/n$$

$$\text{Profit/boarder} = 1600 - a/n - b$$

$$1600 - a/50 - b = 200$$

$$1600 - a/75 - b = 250$$

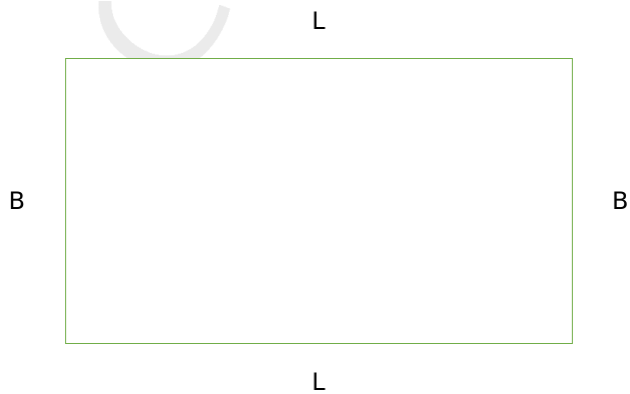
Solving, we get $a = 7500$, and $b = 1250$

Hence, total profit with 80 people = $80 (1600 - 7500/80 - 1250) = 80 (350 - 7500/80) = 28000 - 7500 = \text{Rs. } 20500$

[VIDEO SOLUTION](#)

54.A

Let us draw the rectangle.



Now, definitely, three sides should be fenced at Rs 100/ft, and one side should be fenced at Rs 200/ft.

In this question, we are going to assume that the L is greater than B.

Hence, the one side painted at Rs 200/ft should be B to minimise costs.

Hence, the total cost = $200B + 100B + 100L + 100L = 300B + 200L$

Now, $L \times B = 60000$

$B = 60000/L$

Hence, total cost = $300B + 200L = 18000000/L + 200L$

To minimise this cost, we can use $AM \geq GM$,

$$\frac{\frac{18000000}{L} + 200L}{2} \geq \sqrt{\frac{18000000}{L} \times 200L}$$

$$\frac{18000000}{L} + 200L \geq 2\sqrt{18000000 \times 200}$$

$$\frac{18000000}{L} + 200L \geq 2 \times 60000$$

Hence, minimum cost = Rs 120000.

[VIDEO SOLUTION](#)

55. **3500**

We can say 40m is the perimeter of the park

so side of rhombus = 10

$$\text{Now } \frac{1}{2} \times d_1 \times d_2 = 96$$

$$\text{so we get } d_1 \times d_2 = 192 \quad (1)$$

And as we know diagonals of a rhombus are perpendicular bisectors of each other :

$$\text{so } \frac{d_1^2}{4} + \frac{d_2^2}{4} = 100$$

$$\text{so we get } d_1^2 + d_2^2 = 400 \quad (2)$$

Solving (1) and (2)

$$\text{We get } d_1 = 12 \text{ and } d_2 = 16$$

Now the cost, in INR, of laying electric wires along its two diagonals, at the rate of ₹125 per m, is = $(12+16)(125) = 3500$

[VIDEO SOLUTION](#)

100+ CAT DILR Questions



56. **34**

Let price of smallest cup be $2x$ and medium be $5x$ and large be y

Now by condition 1

$$\text{we get } 2x \times 5x \times y = 800$$

$$\text{we get } x^2 y = 80 \quad (1)$$

Now as per second condition ;

$$(2x + 6) \times (5x + 6) y = 3200 \quad (2)$$

Now dividing (2) and (1)

$$\text{we get } \frac{(2x+6) \times (5x+6)}{x^2} = 40$$

$$\text{we get } 10x^2 + 42x + 36 = 40x^2$$

$$\text{we get } 30x^2 - 42x - 36 = 0$$

$$5x^2 - 7x - 6 = 0$$

we get $x=2$

So $2x=4$ and $5x=10$

Now substituting in (1) we get $y=20$

Now therefore sum = $4+10+20=34$

[VIDEO SOLUTION](#)

57. **8**

Considering the distance travelled by Mira in one minute = M,

The distance traveled by Amal in one minute = A.

Given if they walk in the opposite direction it takes 3 minutes for both of them to meet. Hence $3*(A+M) = C$. (1)

C is the circumference of the circle.

Similarly, it is mentioned that if both of them walk in the same direction Amal completes 3 more rounds than Mira :

Hence $45*(A-M) = 3C$. (2)

Multiplying (1)*15 we have :

$$45A + 45M = 15C.$$

$$45A - 45M = 3C.$$

Adding the two we have $A = \frac{18C}{90}$

Subtracting the two $M = \frac{12C}{90}$

Since Mira travels $\frac{12C}{90}$ in one minute, in one hour she travels : $\frac{12C}{90} \cdot 60 = 8C$

Hence a total of 8 rounds.

Alternatively,

Let the length of track be L

and velocity of Mira be a and Amal be b

Now when they meet after 45 minutes Amal completes 3 more rounds than Mira

so we can say they met for the 3rd time moving in the same direction

so we can say they met for the first time after 15 minutes

So we know Time to meet = Relative distance / Relative velocity

$$\text{so we get } \frac{15}{60} = \frac{L}{a-b} \quad (1)$$

Now When they move in opposite direction

They meet after 3 minutes

$$\text{so we get } \frac{3}{60} = \frac{L}{a+b} \quad (2)$$

Dividing (1) and (2)

$$\text{we get } \frac{(a+b)}{(a-b)} = 5$$

$$\text{or } 4a = 6b$$

$$\text{or } a = 3b/2$$

Now substituting in (1)

we get :

$$\frac{L}{b} \times 2 = \frac{15}{60}$$

$$\text{so } \frac{L}{b} = \frac{1}{8}$$

So we can say 1 round is covered in $\frac{1}{8}$ hours

so in 1-hour total rounds covered = 8.

VIDEO SOLUTION

58. C

Let the alloy contain x Kg silver and y kg copper

Now when mixed with 3Kg Pure silver

$$\text{we get } \frac{(x+3)}{x+y+3} = \frac{9}{10}$$

$$\text{we get } 10x+30 = 9x+9y+27$$

$$9y-x=3 \quad (1)$$

Now as per condition 2

silver in 2nd alloy = $2(0.9) = 1.8$

$$\text{so we get } \frac{(x+1.8)}{x+y+2} = \frac{21}{25}$$

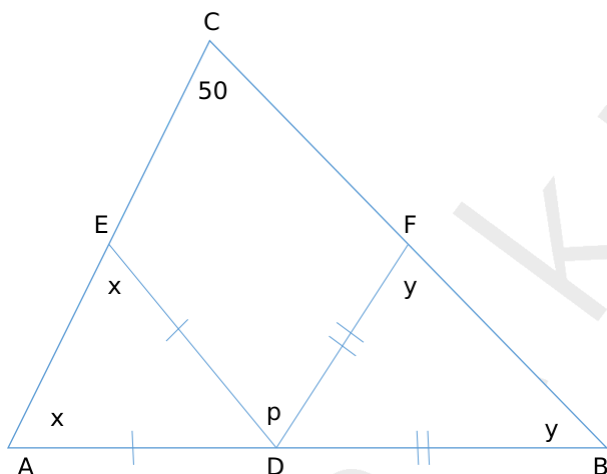
$$\text{we get } 21y-4x=3 \quad (2)$$

solving (1) and (2) we get $y = 0.6$ and $x = 2.4$

so $x+y = 3$

VIDEO SOLUTION

59. B



We need to find out p.

$$\text{Angle ADE} = 180 - 2x$$

$$\text{Angle BDF} = 180 - 2y$$

$$\text{Now, } 180 - 2y + p + 180 - 2x = 180 \text{ [Straight line} = 180 \text{ deg]}$$

$$p = 2x + 2y - 180$$

$$\text{Also, } x + y + 50 = 180 \text{ [Sum of the angles of triangle} = 180]$$

$$x + y = 130$$

$$p = 260 - 180 = 80 \text{ degrees.}$$

[VIDEO SOLUTION](#)

60. B

Bank A: 6% p.a. 1/2 yearly (CI)

Bank B: x% p.a (SI)

Bank C: 2x% p.a (SI)

Let Raju invest Rs P in bank B for t years. Hence, Rupa invests Rs 10,000 in bank C for 2t years.

Now,

$$P \left(\frac{x}{100} \right) t = P \left(1 + \frac{3}{100} \right)^2 - P$$

$$\left(\frac{x}{100} \right) t = 1.0609 - 1$$

$$\left(\frac{x}{100} \right) t = 0.0609$$

We need to calculate

$$\text{SI} = 10000 \times 2t \times \left(\frac{2x}{100} \right) = 40000 \left(\frac{x}{100} \right) t = 40000 \times 0.0609 = 2436$$

[VIDEO SOLUTION](#)

100+ CAT VARC Questions



61. D

Now we have :

$$f(g(x)) - 3x$$

so we get $f(x+3) - 3x$

$$= (x+3)^2 - 7(x+3) - 3x$$

$$= x^2 - 4x - 12$$

$$\text{Now minimum value of expression} = -\frac{D}{4a} = -\frac{(4ac-b^2)}{4a}$$

$$\text{We get } -(16+48)/4$$

$$= -16$$

[VIDEO SOLUTION](#)

62. 3000

Now Anil Paints in 12 Days

Barun paints in 16 Days

Now together Arun, Barun and Chandu painted in 6 Days

Now let total work be W

Now each worked for 6 days

So Anil's work = 0.5W

$$\text{Barun's work} = \frac{6W}{16} = \frac{3W}{8}$$

$$\text{Therefore Charu's work} = \frac{W}{2} - \frac{3W}{8} = \frac{W}{8}$$

$$\text{Therefore proportion of charu} = \frac{24000}{8} = 3,000$$

[VIDEO SOLUTION](#)

63. 6

$$(\sqrt[7]{10})(\sqrt[7]{10})^2 \dots (\sqrt[7]{10})^n > 999$$

$$(\sqrt[7]{10})^{1+2+\dots+n} > 999$$

$$10^{\frac{1+2+\dots+n}{7}} > 999$$

For minimum value of n,

$$\frac{1+2+\dots+n}{7} = 3$$

$$1+2+\dots+n = 21$$

We can see that if $n = 6$, $1+2+3+\dots+6 = 21$.

[VIDEO SOLUTION](#)

64. 50

The question asks for the number of 4 digit numbers using only the digits 1, 2, and 3 such that the digits 2 and 3 appear at least once.

The different possibilities include :

Case 1: The four digits are (2, 2, 2, 3). Since the number 2 is repeated 3 times. The total number of arrangements are :

$$\frac{4!}{3!} = 4.$$

Case 2: The four digits are 2, 2, 3, 3. The total number of four-digit numbers formed using this are :

$$\frac{4!}{2! \cdot 2!} = 6$$

Case 3: The four digits are 2, 3, 3, 3. The number of possible 4 digit numbers are :

$$\frac{4!}{3!} = 4$$

Case4: The four digits are 2, 3, 3, 1. The number of possible 4 digit numbers are :

$$\frac{4!}{2!} = 12$$

Case5: Using the digits 2, 2, 3, 1. The number of possible 4 digit numbers are :

$$\frac{4!}{2!} = 12$$

Case 6: Using the digits 2, 3, 1, 1. The number of possible 4 digit numbers are :

$$\frac{4!}{2!} = 12$$

A total of $12 + 12 + 12 + 4 + 6 + 4 = 50$ possibilities.

Alternatively

We have to form 4 digit numbers using 1,2,3 such that 2,3 appears at least once
So the possible cases :

1	2	3
2	1	1
1	2/1	1/2
0	3/1	1/3
0	2	2

Now we get $\frac{4!}{2!} \times 3 = 36$ (When one digit is used twice and the remaining two once)

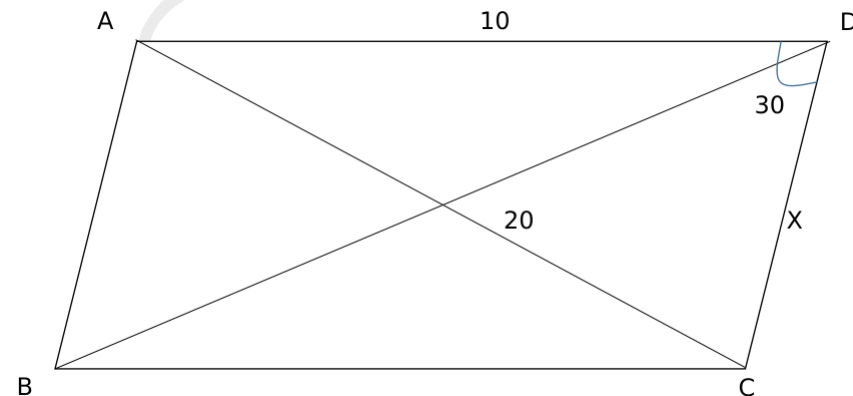
$\frac{4!}{3!} \times 2 = 8$ (When 1 is used 0 times and 2 and 3 is used 3 times or 1 time)

$\frac{4!}{2! \times 2!} = 6$ (When 2 and 3 is used 2 times each)

So total numbers = $36 + 8 + 6 = 50$

[VIDEO SOLUTION](#)

65. B



Applying cosine rule in triangle ACD,

$$100 + X^2 - 2 \times 10 \times X \cos 30 = 400$$

$$X^2 - 10X\sqrt{3} - 300 = 0$$

$$\text{Solving, we get } X = \left(\frac{10\sqrt{3} + 10\sqrt{15}}{2} \right)$$

$$\text{Hence, area} = 10X \sin 30 = \left(\frac{10\sqrt{3} + 10\sqrt{15}}{2} \right) 10$$

$$= 25(\sqrt{3} + \sqrt{15})$$

[VIDEO SOLUTION](#)

CAT Expected Questions PDF



66. C

We need to check for all regions:

$$x \geq 0, y \geq 0$$

$$x \geq 0, y < 0$$

$$x < 0, y \geq 0$$

$$x < 0, y < 0$$

However, once we find out the answer for any one of the regions, we do not need to calculate for other regions since the options suggest that there will be a single answer.

Let us start with $x \geq 0, y \geq 0$,

$$3x + 3y = 7$$

$$2x + 3y = 1$$

Hence, $x = 6$ and $y = -11/3$

Since $y > 0$, this is not satisfying the set of rules.

Next, let us test $x \geq 0, y < 0$,

$$3x - y = 7$$

$$2x + 3y = 1$$

Hence, $y = -1$

$$x = 2.$$

This satisfies both the conditions. Hence, this is the correct point.

WE need the value of $x + 2y$

$$x + 2y = 2 + 2(-1) = 2 - 2 = 0.$$

 VIDEO SOLUTION